

ADVANCING GREEN AVIATION THROUGH COMPOSITE WING DESIGN

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Executive summary

The redesign of the PC-9 aircraft's wing using IM7/977-3 composite materials enhance weight reduction, structural strength, and sustainability. This composite replaces traditional aluminium, achieving a high strength-to-weight ratio with natural resistance to corrosion and fatigue, reducing maintenance and improving fuel efficiency. Static analysis showed the Aluminum wing withstanding operational loads with a 1.5 safety factor, confirming compliance with aviation standards. The wing rib 9 analyzed in the report faces a direct stress of 95.7 mPa approximately and a margin of safety of 3.8.

Dynamic analysis, using safe-life and damage tolerance methods, confirmed durability over extended service. The wing is capable of withstanding 10,000 flights, with specific inspection intervals for high-stress areas, ensuring resilience against fatigue.

The experimental analysis gives us the idea of how wing would react to load cases in real life situations and subsequently we validate our FEA model through this which can be used to design and validate composites. The optimized composite ply configuration : $[-45/0/90/45]_s$ enhances load-bearing capacity, handling multidirectional stresses effectively. We further double the skin thickness for composite spar flange and web to match the stiffness matrix of composite and aluminum.

Life Cycle Assessment (LCA) emphasizes its eco-friendly profile, with lower CO₂ emissions, recyclable options, and efficient manufacturing via Automated Fiber Placement (AFP). These help us understand a systems engineering approach towards internal structure optimization.

Finally experimental validation aligned with FEA, showing consistent stress distributions across the wing. Maintenance improvements include modular design, inspection ports, and real-time load monitoring, ensuring compatibility with legacy systems. This composite redesign elevates the PC-9's structural and environmental performance, setting a sustainable precedent for future aircraft engineering.

Table of Contents

EXECUTIVE SUMMARY	1
TABLE OF TABLES	3
TABLE OF EQUATIONS	ERROR! BOOKMARK NOT DEFINED.
TABLE OF FIGURES.....	4
1. STATEMENT OF PROBLEMS	5
2. INTRODUCTION	6
3. LITERATURE REVIEW	7
4. STATIC ANALYSIS	8
4.1 INTRODUCTION.....	8
4.2 THEORY.....	9
4.3 INPUT DATA	10
4.4 RESULTS.....	23
5 DYNAMIC ANALYSIS.....	25
5.1 INTRODUCTION.....	25
5.2 SAFE LIFE APPROACH:	26
5.3 DAMAGE TOLERANT METHOD	27
5.4 RESULTS.....	30
6 EXPERIMENTAL ANALYSIS	31
6.1 INTRODUCTION:.....	31
6.2 EXPERIMENT SETUP	31
6.3 PROCEDURE:	35
6.4 RESULTS:	35
6. FEA MODELLING	41
INTRODUCTION:	41
THE NUMERICAL METHODS.....	41
FE MODEL DETAILS	42
LOADS AND BOUNDARY CONDITIONS (BCS.....	42
MESH SENSITIVITY ANALYSIS	42
INITIAL MESH GENERATION:	42
VALIDATION OF FE MODEL.....	43
RESULTS	44
DISCUSSION	45
7. COMPOSITES.....	45
7.1 ADVANCEMENTS IN MODERN TECHNOLOGIES	46
7.2 SELECTED MATERIAL	46

7.3 DESIGN SKIN.....	48
7.4 MAIN SPAR DESIGN	51
SUSTAINABILITY AND INTEGRATION	57
CONCLUSION	60
REFERENCES	63
APPENDIX	66

Table of Tables

Table 1: Input Data	10
Table 2 Lift distribution using kutta joukowski theorem.....	11
Table 3 Centroid using first moment of inertia.....	13
Table 4 I section dimensions at rib 9	14
Table 5 Moment of inertias.....	15
Table 6 Notched stresses and Strains.....	17
Table 7 Analysis of Notched stresses and residual stresses after unloading	18
Table 8 Fracture analysis for case 1.....	21
Table 9Material Properties:.....	26
Table 10 PC-9 Loading Spectrum.....	27
Table 11: displacement and load results for each load case	37
Table 12 Mesh sensitive analysis.....	43
Table 13 FEA model validation.....	43
Table 14 Three cases result for each parameter	44
Table 15 : IM7/977-3 material properties	48
Based on the FEA of wing under design ultimate load cases, we find the stresses in each direction to be positive hence we will use tensile strain to calculate the strengths in each direction.	49
Table 17 stress values in each direction (DUL).....	49
Table load intensity in each direction	50
Table 19 Strength in each orientation	50
Table Skin design.....	51
Table number of plies at rib 5.....	52
Table Em Eb values for composite (IM77/973)	52
Table 23 Axial and bending stiffnesses for both Aluminium and Composites	52
Table New dimensions based on Axial stiffness	53
Table Stiffness comparison after iteration.	53

Table of figures

Figure 1 wing main spar	8
Figure 2: Aerodynamic forces and moments (www.sciencedirect.com, n.d.)	10
Figure 3 Span wise lift distribution.....	12
Figure 4: Spar dimensions	14
Figure 5 Unnotched Stress Analysis of upper half of Spar.....	15
Figure 6 Peterson's Stress Concentration curve	16
Figure 7 Cases of cracks on the Spar	19
Figure 8 crack emanating from hole	20
Figure 9 a/b vs F curves for different crack types	20
Figure 10 Fracture analysis for case 2	22
Figure 11 Fracture Analysis of case 3.....	23
Figure 12 Wooden clamping on wing with two actuators below at rib 11	32
Figure 13 Measurement tape for displacement.....	32
Figure 14: position of sensors	33
Figure 15 strain gauge sensor	33
Figure 16 Rib-6	33
Figure 17 Rib-9 (MLG-X)	34
Figure 18 Displacement, load, stress and strain values will be visible on screen as in above picture	34
Figure 19: wing displacement at load (mm) over time(s).....	35
Figure 20: force (N) over displacement (mm)	36
Figure 21: comparison of wing tip displacement and displacement at load overload	36
Figure 22 FE analysis for aluminium.....	44

1. Statement of Problems

The aviation industry is increasingly prioritizing sustainable, eco-friendly practices to mitigate carbon emissions and reduce resource consumption. In this context, PAL, a global aircraft manufacturer, is aiming to advance the sustainability of its PC-9 aircraft by redesigning its wing with innovative composite materials and structural solutions. This project seeks to address challenges related to reducing the carbon footprint, enhancing structural efficiency, managing aeroelastic behaviour, improving fatigue resistance, validating experimental data, and ensuring manufacturability.

To achieve these goals, PAL will focus on:

- 1. Employing advanced materials and manufacturing techniques:** This includes integrating traditional, reliable components with vacuum-formed composites, which are now more affordable and offer enhanced directional strength due to fibre orientation.
- 2. Applying advanced design methodologies:** Through structural analysis and materials science, this effort seeks to optimize the aircraft's weight and drag, boost fuel efficiency, reduce noise, and enable reliable operation across various missions.
- 3. Utilizing enhanced design simulations (FEM):** Modelling and simulating various material and component configurations under realistic constraints to predict performance outcomes.
- 4. Integrating advanced systems:** Exploring sustainable options such as hydrogen fuel and other eco-friendly alternatives, alongside adaptive systems, electrical configurations, and intelligent technologies to improve aircraft performance and efficiency.
- 5. Implementing innovative design configurations:** Developing a forward-looking wing structure for fixed-wing aircraft.

The successful redesign and implementation of a composite wing for the PC-9 will not only reduce its carbon footprint but also set a foundation for more environmentally conscious aircraft designs in the future. By leveraging finite element analysis (FEM), the project aims to assess the feasibility of these advanced technologies in enhancing the PC-9's performance and aligning with green aviation advancements.

2. Introduction

This report will focus on the study, improvement, and redesign of the Pilatus Aircraft Manufacturing Company's PC-9 trainer aircraft. Majorly focusing on comprehensive study and redesign of the PC-9 aircraft wing, with the aim of transitioning from a traditional aluminium structure to an optimized composite design. The initial phase of the study examines the current wing structure made from Al2124 T851 aluminium alloy, exploring the material's performance and its limitations under various load conditions. By redesigning the wing with advanced composite materials, the project aims to enhance the wing's performance, reduce weight, and support contemporary sustainability goals in aerospace.

The report begins with an analysis of lift distribution across the existing wing structure, assuming an elliptical lift distribution to simulate a typical aircraft lift profile. Understanding this distribution is essential for accurate load calculations and structural analysis, as it directly impacts the stresses and forces experienced by the wing during flight. Following this, a static analysis is conducted to determine the loads acting on the wing and the resulting stresses in both the x- and y-directions. Calculations of principal stresses, von Mises stresses, and margins of safety are included to evaluate the wing's structural integrity and its ability to withstand operational stresses without failure.

Further, the report delves into the analysis of notched and residual stresses. The presence of notches or discontinuities in the material can lead to localized stress concentrations, potentially compromising structural integrity. The study calculates these stress concentrations and examines the deformation and residual stresses present after unloading, providing insight into how the material reacts under cyclic loading. Additionally, a fracture analysis assesses crack propagation and determines the margin of safety against brittle fracture and plastic collapse in three specific cases. This analysis is critical in evaluating the wing's durability and resistance to crack growth, especially under fluctuating load conditions.

The study then proceeds with a dynamic analysis, employing both safe-life and damage-tolerance methods to evaluate fatigue resistance. For this analysis, the wing's web is assumed to bear shear forces with a constant load, and the effects of environmental factors on load distribution are acknowledged. While mathematical models provide a robust understanding of structural behaviour, certain environmental conditions cannot be fully accounted for analytically, introducing minor errors into the final results.

Following the analytical methods, experimental analysis is conducted to measure strains at various points across the wing structure. This experimental data is validated through finite element analysis (FEA), which provides a detailed simulation of the wing's behaviour under load. To optimize the redesigned composite wing, a model analysis is conducted under lift and drag loads to derive the maximum stresses in the x-, y-, and xy-directions. These maximum stresses are then compared with the strengths of potential composite materials in each direction, allowing for the selection of a composite with higher strength in each orientation. This iterative approach ensures the composite skin can withstand 1.5 times the load experienced by the original aluminium skin, providing an added safety margin.

Finally, the report presents the design of a composite wing spar, which achieves significant weight savings while meeting all required strength and durability criteria. The composite wing design is validated through finite element analysis, confirming its suitability as a lightweight and efficient alternative to the aluminium structure. This analysis represents a step forward in

the transition to sustainable materials and advanced design practices, aligning with the aerospace industry's push toward environmentally conscious innovations.

3. Literature Review

The PC-9/A aircraft, initially designed as an advanced variant of the PC-7, became a full replacement due to the integration of the Electronic Flight Information System (EFIS), also known as the “glass cockpit.” This two-seat, single-engine turboprop trainer aircraft was manufactured by the Swiss company Pilatus. The Australian Air Force ordered the PC-9/A after cancelling their A10 Wamira trainer order in December 1985, drawn by the PC-9/A's more powerful engine and larger cockpit. Production commenced that year, with the first flights occurring in December 1987. Deliveries to the Royal Australian Air Force (RAAF) spanned from 1987 until March 1992. The PC-9/A was later succeeded by the PC-21 in 2019. With a length of 10.14 meters, wingspan of 10.125 meters, and height of 3.26 meters, the PC-9 features a maximum take-off weight of 3,200 kg and a Pratt & Whitney PT6A-62 engine delivering 708 kW of power. With a cruising speed of 556 km/h and a maximum range of 1,537 km, it served effectively as a versatile trainer during the 1990s.

The internal structure of the PC-9 wing includes 11 main ribs riveted to two central spars and five stringers on both the upper and lower surfaces for additional support. Besides these primary components, additional structural elements support the ailerons, flaps, and fuel tank. This project aims to enhance the PC-9's efficiency by utilizing advanced materials technology, potentially allowing for a reduction in supporting structures and resulting in both cost and weight savings.

Static Analysis of Wing (Literature Review):

Static analysis enables theoretical calculations of the stresses in the wing spar, providing a basis for redesign. According to Blasi et al. (2007), wing behaviour depends on factors such as mass distribution, material elasticity, aerodynamic forces, and moments. Using theoretical formulas and reasonable assumptions, we can assess force distribution, including lift, self-weight, and the stresses in the internal structure. Computational advancements now allow automated wing analysis, with software like XFOIL and XFLR5 capable of calculating lift and lift coefficients for both 2D airfoil sections and 3D wing models. Xie et al. (2013) noted that traditional static analysis can be inaccurate for large wing deformations, advocating nonlinear static aeroelastic and trim analysis for more precise results that align closely with experimental data.

Dynamic Analysis of Wing (Literature Review):

Initially, aircraft structural design focused primarily on strength, but over time researchers recognized that fatigue and corrosion significantly impact performance as well. Due to fluctuating wing loads, aircraft wings may experience failure in certain conditions, with stress concentrations leading to fatigue cracking in structural areas. Recently, dynamic research has advanced significantly with simulation tools like ANSYS. Pandey et al. (2021) used ANSYS to simulate an aircraft wing with embedded cracks, applying lift and drag forces to calculate the stress intensity factor and energy release rate, enabling predictions of fatigue life under varying loads. Additionally, Li et al. (2017) developed probabilistic models using dynamic Bayesian networks to forecast future crack growth in wings, enhancing fatigue analysis. Although these methods improve fatigue life prediction, occasional fatigue-related issues persist. This project used the PC-9's technical specifications and G-A-G spectrum data to evaluate fatigue life based on the principles of safe-life and damage tolerance design.

4. STATIC ANALYSIS

4.1 INTRODUCTION

In static analysis of a wing, understanding the lift distribution is essential as it directly influences the internal load distribution and structural integrity. Lift distribution determines how forces are spread along the wingspan, affecting the wing's bending moments and shear forces. This distribution is crucial because non-uniform lift results in varying stresses, which are critical in predicting maximum stress locations and potential failure points. Additionally, lift distribution impacts wing deflection, as a non-optimized load distribution can lead to excessive bending or twisting, compromising both aerodynamic performance and safety. Proper analysis enables designers to optimize structural elements, such as spars and ribs, to handle the stresses efficiently and minimize weight. Furthermore, variations in lift distribution contribute to fatigue over the aircraft's lifespan, making it imperative to have accurate lift models for long-term durability predictions. Therefore, lift distribution forms the backbone of structural optimization, ensuring that the wing can withstand operational loads while maintaining a lightweight and efficient design.

The scope of static analysis is to understand the response of the wing to aerodynamic forces and the moments caused due to them. It involves a comprehensive evaluation of structural integrity under various loading conditions to ensure safety and performance. This analysis begins with assessing the loads acting on the wing, including lift, weight, and aerodynamic forces during cruise. Key components of static analysis include calculating limit and ultimate loads, where limit loads represent maximum expected conditions and ultimate loads incorporate a safety factor, typically 1.5, to account for uncertainties in materials and manufacturing (FAA, 2016). Engineers analyse vertical shear forces and bending moments resulting from the lift distribution, particularly at critical locations such as the wing root, to design reinforcement and ensure the structure can withstand these stresses without permanent deformation. Additionally, static analysis must consider torsional and horizontal shear loads, which further influence overall structural performance. By employing methods such as shear and bending moment diagrams, static analysis validates that the wing can safely endure the anticipated loads throughout its operational envelope, ultimately ensuring the aircraft's reliability and compliance with regulatory standards.

In our project we will observe the forces and moments and the stresses at rib 9 of the PC-9 wing along the main spar. As presented below.

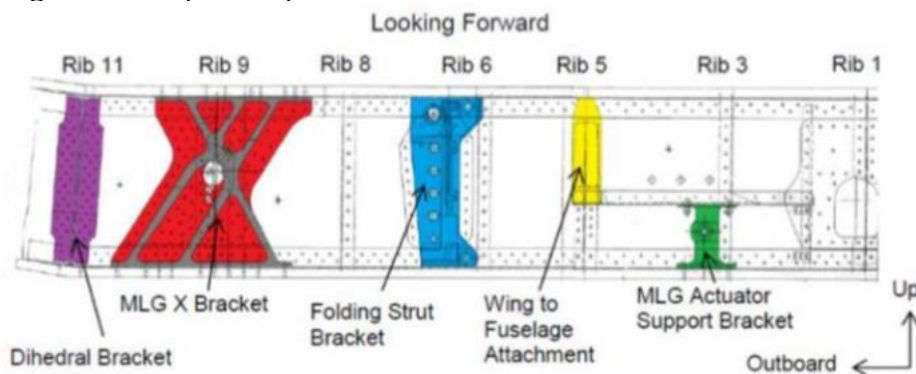


Figure 1 wing main spar

4.2 THEORY

In order to determine the lift distribution over the wing we incorporate Prandtl's lifting line theory and the Kutta Joukowski theorem to assume an elliptical distribution of lift over the wing from root to tip. Prandtl's theory assumes wing as a series of lift generating vortices (Cheng and Wang 2018) despite the complexities of 3-dimensional flow. We then use the Kutta Joukowski theorem that makes use of vortices to produce spanwise lift distribution, incorporating circulation which then helps us estimate elliptical lift distribution along spanwise wing length. Considering this as a preliminary analysis we can assume the flow over wing is compressible and inviscid (NASA 1982).

We start by calculating the weight of the aircraft at cruise assuming negligible loss of weight due to fuel consumption hence we assume at cruise, lift provided is equivalent to the weight, another assumption is of the consideration of the downforce provided by tail, being small so we neglect it (THG Megson 2017).

Equation 1:

$$L = W = MTOW * g$$

The MTOW is the maximum take-off weight of the aircraft which is used to determine the magnitude of lift in cruise. this forms as our basis for the use of the elliptical lift distribution theorem where we use the lift which acts at centreline to estimate distribution across the wing span (Fluids -Lecture 7 Notes Elliptical Lift Distribution, n.d.).

We know,

Equation 2:

$$L = \frac{\pi}{4} * \rho * V\Gamma_0 * b$$

From equation 1 we know the lift and hence we calculate Γ_0 which is the circulation at the centerline of the aircraft fuselage. Hence,

Equation 3:

$$\Gamma_0 = \frac{4 * L}{\rho * V * b}$$

Elliptical distribution of circulation Γ_0 along wing span can further be provided using,

Equation 4:

$$\Gamma_y = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2}$$

Where y is an arbitrary span wise position and b is the wing span. Hence, we can finally derive lift per span using equation 2 and substituting Γ_y instead of Γ_0 and integrate it along the required span.

Equation 5:

$$\int_{-\frac{b}{2}}^{\frac{b}{2}} \rho * V * \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} dy$$

We will be calculating loads and moments about rib 9 hence we will modify equation 5 as per required between rib 9 and wing tip and use the equation to find concentrated load and the find its distance from the rib 9 using Area moment of inertia to find centroid.

Following this we will apply basics of flight mechanics to estimate forces and moments acting on the wing based off the action of lift along chordwise position and its offset from the main spar.

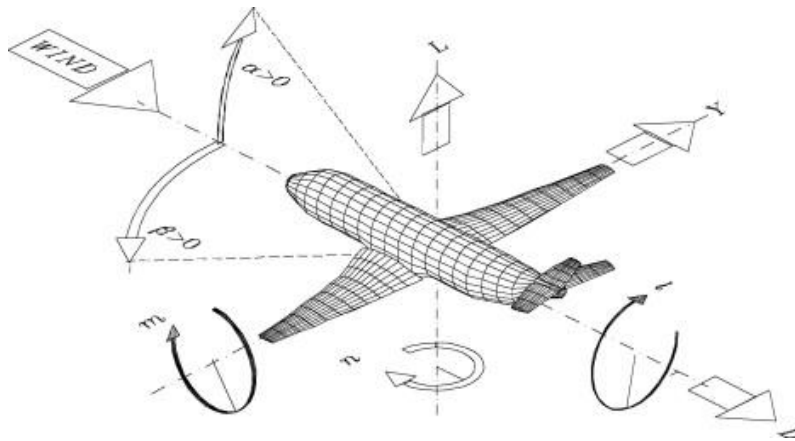


Figure 2: Aerodynamic forces and moments (www.sciencedirect.com, n.d.)

The above figure gives us a broad idea of how the forces and moments act around the aircraft in cruise. Our co-ordinate system follows Lift (L) along positive Z axis (upward), Drag (D) along negative x axis (rearward fuselage) and span along Y axis, with clockwise moment positive and left side is positive Y axis, compression as negative and tension positive.

Major Assumptions used:

1. Lift acts along centre of pressure at 25% mean aerodynamic chord MAC.
2. Flanges carry bending and web carries shear and is effective in fully resisting direct stresses.
3. Compression retards crack growth and tension accelerates it.
4. ribs before rib 5 are clamped and
5. safety factor of 1.5 (ASTM 36, FAR 23)

4.3 Input Data

M (kg)	3200	Maximum take-off weight
rho(kg/m ³)	0.572	density
v(m/s)	154.4	freestream velocity
b(m)	10.124	wingspan
S(m ²)	16.29	wing area
MAC(m)	1.65	mean aerodynamic chord
AR	6.3	Aspect ratio

Table 1: Input Data

Using the above input data, we find the following

$$L = W = MTOW * g$$

$$L = 3200 \times 9.8 = 31360 \text{ N}$$

$$L = W * 1.5 = 31360 * 1.5$$

Hence required lift after consideration of the safety factor,

$$L = 47,040 \text{ N}$$

This helps us factor in loads that are beyond the scope of static analysis and assume a broader safety margin. The downside to this is we will result in a lowered margin of safety upon calculation, which as explained above will factor in extreme cases, hence those will be considered safe.

From equation 3,

$$\Gamma_0 = \frac{4 * L}{\rho * V * b}$$

$$\Gamma_0 = \frac{4 * 47,040}{0.572 * 154.4 * 10.124}$$

Hence

$$\Gamma_0 = 67.0196561 \text{ m/s}^2$$

from this we determine the circulation and the following lift distribution along the span as presented below using equations 4&5

example for rib 9, $y = 1.044 \text{ m}$

hence,

$$\Gamma_y = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2}$$

$$\Gamma_y = 67 \sqrt{1 - \left(\frac{2 * 1.044}{10.124}\right)^2} = 65.57 \text{ m/s}^2$$

$$L_y = \rho * V * \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2}$$

$$L_y = 5791 \frac{\text{N}}{\text{m}} \text{ (here the } L_y \text{ is lift per unit span)}$$

hence, we calculate the same for every rib along the span and also plot it to explain the application of the theorem. a major limitation here is that this theorem only applies for finite wing along the span and is in applicable if we were to find pressure distribution along chordwise position. but to account for those we will calculate all the possible forces and moments that can be observed due to the application of lift at 25% MAC.

rib	y(m)	Ty	Ly(N/m)
1	0	67.0196561	5918.96156
3	0.23	66.9504399	5912.84861
5	0.46	66.7423605	5894.47171
6	0.633	66.4935865	5872.50078
8	0.846	66.0770428	5835.71298
9	1.044	65.5787932	5791.70916
11	1.25	64.9441438	5735.65896
12	1.51	63.9683738	5649.48207
13	1.77	62.7890477	5545.32777
14	2.03	61.3944363	5422.16015
15	2.29	59.7694716	5278.64847
16	2.55	57.8947605	5113.07998
17	2.81	55.7451117	4923.22988
18	3.085	53.1351911	4692.73005
19	3.09	53.0842389	4688.23011
20	3.36	50.1265079	4427.01278
21	3.635	46.6419582	4119.2685
22	3.91	42.5648323	3759.18978
23	4.185	37.7033795	3329.84183
24	4.46	31.6987684	2799.53379
25	4.735	23.6973883	2092.8775
26	5.01	9.58163415	846.219267

Table 2 Lift distribution using kutta joukowski theorem

The above table provides us a distribution over the wing span, and can be further plot to explain the elliptical lift distribution.

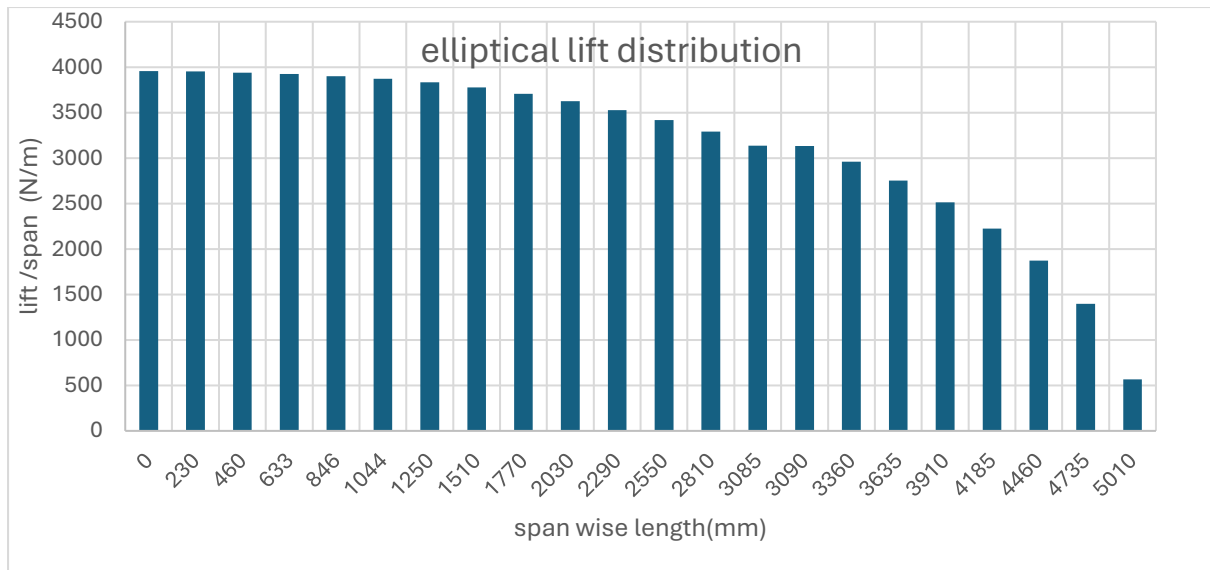


Figure 3 Span wise lift distribution

once we have the distributed load our next step is to convert that into concentrated load, acting at a point along the spanwise location.

using equation 5 and modifying it to fit our parameters,

$$\int_{1.044}^{5.01} \rho * V * \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} dy$$

$$\int_{1.044}^{5.01} 0.572 * 154.4 * 67 \sqrt{1 - \left(\frac{2y}{b}\right)^2} dy$$

$$L_{concentrated} = 17329.9 \text{ N}$$

this is understandable as half wing incorporates half the weight if simply calculated, but since our approach incorporates for wing span distribution based on circulation and lift at centreline we can safely use the weight of the aircraft that obeys equilibrium. our next step would be to find the section centroid to determine where the lift would act, for this we use the first moment of area as we would for any other shape. another downside to using elliptical model is the estimated location of point where the concentrated load acts.

Using first moment of area we divide the ellipse from rib 9 to wing tip into arbitrary elements that are rectangular in shape with each rectangle consisting of a vertical element that represents the load distribution and horizontal as its breadth.

example at rib 9 we would have the height as L_y , breadth as y .

$$L_y = 5791 \frac{\text{N}}{\text{m}}$$

$$y_i = y_n - y_{n-1} = y_{rib9} - y_{rib8}$$

$$y_i = 1.044 - 0.846 = 0.198 \text{ m}$$

$$A_i = L_y * y_i = 1146.75 \text{ N}$$

$$A_i * y_i = 1146.7 * 0 = 0 \text{ (we start our shape at rib 9)}$$

$$\text{Equation 6: } y_c = \frac{\sum A_i * y_i}{\sum A_i}$$

hence provided below is the table of first moment of inertia distribution along the span from rib 9 to rib 25 of left-hand wing

rib	y(m)	Ty	Ly(N/m)	Ai	AiXi
9	0	65.5787932	5791.70916	1146.75841	0
11	0.206	64.9441438	5735.65896	1181.54574	243.398423
12	0.466	63.9683738	5649.48207	1468.86534	684.491248
13	0.726	62.7890477	5545.32777	1441.78522	1046.73607
14	0.986	61.3944363	5422.16015	1409.76164	1390.02498
15	1.246	59.7694716	5278.64847	1372.4486	1710.07096
16	1.506	57.8947605	5113.07998	1329.4008	2002.0776
17	1.766	55.7451117	4923.22988	1280.03977	2260.55023
18	2.041	53.1351911	4692.73005	1290.50076	2633.91206
19	2.046	53.0842389	4688.23011	23.4411506	47.960594
20	2.316	50.1265079	4427.01278	1195.29345	2768.29963
21	2.591	46.6419582	4119.2685	1132.79884	2935.08179
22	2.866	42.5648323	3759.18978	1033.77719	2962.80542
23	3.141	37.7033795	3329.84183	915.706503	2876.23413
24	3.416	31.6987684	2799.53379	769.871791	2629.88204
25	3.691	23.6973883	2092.8775	575.541313	2124.32299
26	3.966	9.58163415	846.219267	232.710298	922.929044
			sum	17800.2468	28315.8481
centroid using first moment of area					
cp		sum(Aixi)/sum(Ai)		1.59075593	

Table 3 Centroid using first moment of inertia

hence our concentrated point (cp) or y_c is at 1.6 m from rib 9.

we now have necessary parameters to determine forces and moments around the rib 9.
given the above data, we see that the lift, $L_{concentrated}$ produces a counter clockwise moment M_x (about X axis namely rolling moment) and a moment M_y (pitching moment) which acts due to offset from rib 9 which lies along main spar at 35% chord wise position which is clockwise direction and causes a positive moment. there is a third moment around Z axis due to drag but in cruise (trim) state, those forces are usually minimum and can be ignored due to negligible effect as no side slip angle is mentioned. hence assuming $\beta = 0$

$$M_x = L_{concentrated} * y_c = 17329 * 1.6(m) = -27549467 \text{ Nmm (we convert to Nmm)}$$

$$M_y = L_{concentrated} * 0.1 * MAC = 17329 * 0.16(m) = 2857548.38 \text{ Nmm}$$

Drag (D),

$$\text{Equation 7: } D_{rib9} = \frac{1}{2} * \rho * v^2 * S * \left(\frac{Cl^2}{\pi * Ar} + C_{di} \right)$$

$$Cl = 2 * \frac{L}{\rho * v^2 * S} = 0.15592937$$

$c_{di} = 0.011$ (estimated from drag polar)

hence,

$$D_{rib9} = \frac{1}{2} * 0.572 * 154.4^2 * 16.29 * \left(\frac{0.156^2}{\pi * 6.3} + 0.011 \right) = 1358.41385 \text{ N}$$

Moments of inertias:

$$\frac{bh^3}{12} I_{xx} = \frac{bh^3}{12} + Ay^2 \quad \text{Equation 8 (parallel axis theorem)}$$

$$I_{zz} = \frac{b^3 * h}{12} \quad \text{Equation 9 (parallel axis theorem)}$$

$$I_{zz} = \frac{b^3 * h}{12} + Ax^2 \quad \text{Equation 10 (parallel axis theorem)}$$

root		
I-Beam	mm	
tw	2.3	
tf	17	
hw	226	
bf	70	
A_flange(mm2)	1190	mm2
A_web(mm2)	519.8	mm2

Table 4 I section dimensions at rib 9

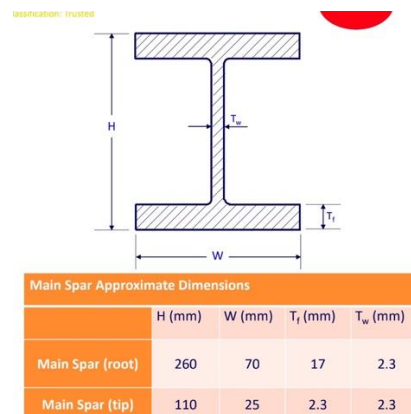


Figure 4: Spar dimensions

Hence, we can use flexural formula to calculate the bending stresses in X and Y and further calculate shear stresses.

Equation 11

$$\sigma_y = \frac{Mx * z}{I_{xx}}$$

Equation 12

$$\sigma_y = \frac{Mx * z}{I_{xx}}$$

we also factor for the shear stress caused by drag due to Torsion,

Equation 13

$$T = D_{rib9n} * C$$

where c is the distance from torsion center that is the centroid of, I beam due to symmetry. A major consideration with respect to the co-ordinate system, hence the cross section lies in X-Z plane and Y is perpendicular out of plane. hence applying perpendicular axis theorem,

Equation 14

$$I_{yy} = I_{xx} + I_{zz}$$

To account for torsion constant J, we will simply calculate using sum of polar moments of inertias from each rectangle along the I section,

Equation 15:

$$J = 2 * \frac{bh^3}{3} \left(1 - \frac{0.21h}{b}\right) + \frac{bh^3}{3} \left(1 - \frac{0.63h}{b}\right)$$

and the shear stress due to Torsion,

Equation 16

$$\tau = \frac{Tr}{j}$$

Web is effective in resisting bending, shear in web can be given by following equation:

Equation 17

$$q = -\frac{S}{I_{xx}} * \left(\int_0^s t_d * y ds + A_{flange} * d \right)$$

here S is the shear force through web which is equal to the concentrated load, hence

$$q = -\frac{17329}{37346597.1} * \left(\int_0^{113} 2.3 * y ds + 70 * 17 * 121.5 \right)$$

here d is the distance from centroid to flange.

Equation 18

$$\tau = \frac{q}{t}$$

where it is the thickness of web at point B and C

Due to symmetry, we analyse three points, A, B and C where A lies on centreline of top flange and B lies on flange web junction and C at section centroid.

Table 5 Moment of inertias

Ixx(mm4)	37346597.1
Iyy(mm4)	38318430.4
Izz(mm4)	971833.333
J(mm4)	9010608.33

Equation 19:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + 4\tau^2}$$

Equation 20:

$$\sigma_{vonmis} = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2}$$

This gives us the necessary stresses to calculate for principal stresses, Von missus stresses and Margin of Safety. Hence using Equations 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 we present below the stress analysis:

location	point	x(mm)	z(mm)	σ_x	σ_y	q	τ	σ_1	σ_2	Von Mis	MoS	Type
top of flange	A	0	130	-9.6945852	-95.897109	0	2.54779625	-9.6193484	-95.972346	91.5425123	3.81743388	compression
web flange ju	B	-1.15	113	-8.4268318	-83.356718	-73.85685639	-32.111677	3.45173277	-95.235282	97.0072171	3.54605351	compression
centroid	C	0	0	0	0	-49.23790426	-21.407784	21.4077845	-21.407784	37.0793704	10.8934058	compression

Figure 5 Unnotched Stress Analysis of upper half of Spar

Following the results from Unnotched stress analysis, we proceed to calculate the stresses around notches, which indicate the stress levels around holes in the Spar. for our analysis we overlook at cases of two types as shown:

1. Holes in flanges.
2. Holes in web.

For hole with diameter of 6.35mm. As observed from the Kt curve from Peterson's Stress Concentration Factors Handbook, $a=b$ and hence for $a/b=1$ Kt yields a factor of 4 for all cases as it is not specified whether the notch is perfectly circular or no.

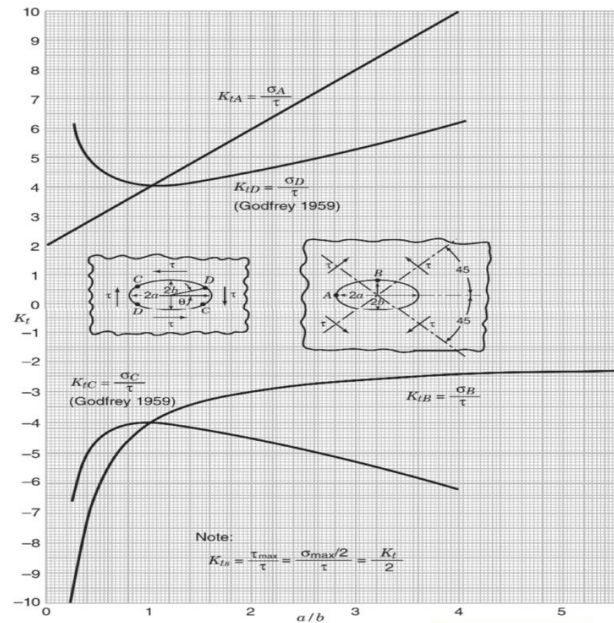


Figure 6 Peterson's Stress Concentration curve

Kt=4 approximately.

For the flange, nominal stress is just the stress perpendicular to the flange cross section that is $S = \sigma_y = 95.9 \text{ mPa}$ (compression for top tension for bottom)

$$\begin{aligned} \text{Equation 21} \\ \sigma = Kt * S \end{aligned}$$

Before conducting the notched stress analysis, a major step is to check for plasticity,

$$Kt * s > \sigma_0$$

in our case $Kt * S = 383.6 \text{ mPa} < 441 \text{ mPa}$ which indicates the notched stresses are elastic in this case we simply use hook's law and calculate stresses and strains from equation 20, $\sigma_{notched} = 383.1 \text{ mPa}$

According to Hook's law,

$$\begin{aligned} \text{Equation 22} \\ \varepsilon = \sigma / E \end{aligned}$$

Hence the notched strain,

$$\varepsilon = \frac{383.6}{73100} = 0.005248 \frac{\text{mm}}{\text{mm}}$$

For the web,

Our assumption being web is only active in shear, we calculate the total load acting on the spar as,

Equation 23

$$P_y = \frac{Mx}{H_{spar}}$$

$$P_y = \frac{27549467}{260} = 105758.379$$

Equivalent stress on web is,

$$\sigma_{web} = \frac{P_y}{A} = \frac{105.76kN}{(226 - 6.35) * 2.3} = 209.3mPa$$

Hence from figure 6 it can be approximated that Kt for web is around 4
We have assumed web is active in shear, hence we will use the equation 20 and check the validity against the shear strength in this case.

Hence, we check for the plasticity condition,

$$Kt * S$$

Where $Kt * S = 837.36mPa \text{ (compression)} > 295mPa \text{ (plastic)}$

Hence the notched stresses in web are plastic. Hence, we use Neuber's equation and Ramberg Osgood model to determine the notched stress and strain value

From Neuber's equation:

Equation 24

$$\sigma \varepsilon = \frac{(KtS)^2}{E}$$

$$\sigma \varepsilon = \frac{(KtS)^2}{E} = \frac{(4 * 209.34)^2}{73100} = 9.5920$$

Given Ramberg Osgood relation:

Equation 25

$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{H}\right)^{\frac{1}{n}}$$

Here H=591mPa and n=0.053,

Hence from equation 23 and 24 we can solve for notched stresses and strain as:

$$\sigma = 480.3 \text{ mPa (compressive)}$$

$$\varepsilon = 0.019971 \frac{mm}{mm}$$

Hence the final values for both web and flange can be computed in the table below:

Table 6 Notched stresses and Strains

Analysis around notches		
Section	$\sigma(mPa)$	$\varepsilon(mm/mm)$
Flange	383.6	0.05248
Web	480.3	0.019971

When the Aircraft is on ground conditions, the stresses go back to zero causing residual stresses and strains,

$$\Delta S = S - S_0$$

Where

$$S_0 = 0(\text{ground conditions})$$

Hence, we repeat the calculations for checking for plasticity,

$$Kt * \Delta S > 2\sigma_0$$

For flange we find that:

$$Kt * \Delta S = 386.3 < 882 \text{ mPa}$$

Hence upon unloading there is no plastic deformation on unloading.

$$\text{Equation 26}$$

$$\Delta\sigma = Kt * \Delta S$$

$$\Delta\varepsilon = \frac{\Delta\sigma}{E}$$

Hence, we find residual stresses as:

$$\text{Equation 27}$$

$$\sigma_r = \sigma_{\text{notched}} - \Delta\sigma$$

$$\varepsilon_r = \varepsilon_{\text{notched}} - \Delta\varepsilon$$

Hence from equation 25 and 26 we can calculate

Residual stresses and strains for flanges are zero as there is no plastic deformation found on unloading.

For web acting in shear,

$$Kt * \Delta S = 4 * 209.34 = 837.36 > 590 \text{ (shear strength of Al2124-T851 is 295mPa)}$$

Hence plastic deformation found on unloading

Hence, we will use modified Neubers rule by replacing ε, σ with $\frac{\Delta\varepsilon}{2}$ & $\frac{\Delta\sigma}{2}$ in the Ramberg Osgood relation (equation 23 and 24)

Hence

$$\Delta\sigma\Delta\varepsilon = \frac{(Kt\Delta S)^2}{E}$$

$$\Delta\sigma\Delta\varepsilon = \frac{(Kt\Delta S)^2}{E} = \frac{(4 * 209.34)^2}{73100} = 9.5920$$

$$\frac{\Delta\varepsilon}{2} = \frac{\frac{\Delta\sigma}{2}}{E} + \left(\frac{\frac{\Delta\sigma}{2}}{H} \right)^{\frac{1}{n}}$$

$$\text{Hence } \sigma_r = -415.55 \text{ mPa} \text{ \& } \Delta\varepsilon = 0.0077158 \frac{\text{mm}}{\text{mm}}$$

Hence, we can compute the results below:

Table 7 Analysis of Notched stresses and residual stresses after unloading

Notched stresses			Residual stresses	
Section	$\sigma(\text{mPa})$	$\varepsilon(\text{mm/mm})$	$\sigma_r(\text{mPa})$	$\varepsilon_r(\text{mm/mm})$
Flange	383.6	0.05348	0	0
Web	480.3	0.019971	-415.55	0.0077158

Analysis of stresses around notches is followed by Fracture Analysis. In Fracture Analysis we observe 3 cases:

1. Cracks emanating from hole on the flanges
2. Cracks on side of flanges

3. Centre crack on web

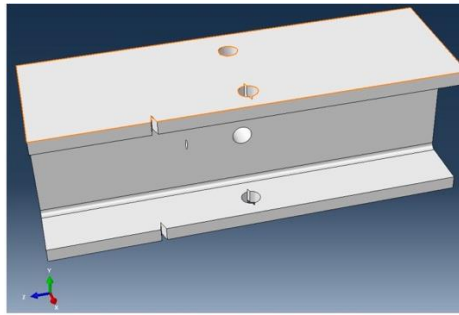


Figure 7 Cases of cracks on the Spar

For cracks emanating from hole on Top flange, from our results in unnotched analysis, we can see that the top flange faces compression hence it can be seen that the crack growth retards under compression while there is a minor shear stress on the flange of about 2.5mPa, we can assume that the crack growth under shear loading to be negligible hence high MoS against brittle fracture and plastic collapse.

Crack growth under shear loading, commonly referred to as Mode II, is characterized by sliding motion along the crack faces due to applied in-plane shear stresses. The propagation of a crack in this mode is primarily influenced by the stress intensity factor (SIF), denoted as K_{II} , which quantifies the intensity of shear stress at the crack tip. For an infinite body under uniform shear stress, the stress intensity factor can be calculated using the formula

Equation 28

$$K_{II} = \tau * \sqrt{\pi * a_i}$$

where τ represents the applied shear stress and a_i is the crack length (Anderson, 2005). This formula highlights how increasing shear stress or crack length escalates the likelihood of crack propagation under Mode II conditions.

Additionally, when both Mode I (tensile) and Mode II (shear) loadings are present, the crack growth direction and rate become more complex, and the effective stress intensity factor can be calculated as

Equation 29

$$K_{effective} = \sqrt{K_I^2 + K_{II}^2}$$

Where K_I and K_{II} are the SIFs for Mode I and Mode II, respectively (Broek, 1986). This equation accounts for the combined effects of tensile and shear loading on the crack. Crack propagation begins when K_{II} reaches the critical value, K_{IIc} , the material's fracture toughness in Mode II, indicating the point at which crack growth becomes unstable and rapid (Suresh, 1991).

The assumption that the Mode II fracture toughness K_{IIc} is approximately 1.5 times the Mode I fracture toughness K_{Ic} for certain materials, including ductile metals like aluminium alloys, is based on experimental observations and general fracture mechanics principles. This relationship is not universally applicable but is often used as a rough estimate in various sources.

For example, Richard and Fulland (1995) suggest that for many ductile materials, the Mode II toughness can be significantly higher than the Mode I toughness due to the different stress and strain fields generated at the crack tip under shear loading. Similarly, Murakami (1987) highlights that the larger plastic zone formed during shear (Mode II) loading generally results in higher energy dissipation before crack propagation, which leads to higher KIIC values compared to KIC.

This rule of thumb, however, should not replace experimental data, as the precise ratio between KIIC and KIC is material-specific and depends on factors such as alloy composition and microstructural characteristics.

Hence, we will calculate our fracture toughness's and margins of safety for tensile loads as those are more significant in crack propagation for the three cases mentioned above.

Critical mode 1 fracture toughness for our material is $26\text{MPa}\sqrt{\text{m}}$ and yield strength of 441MPa .

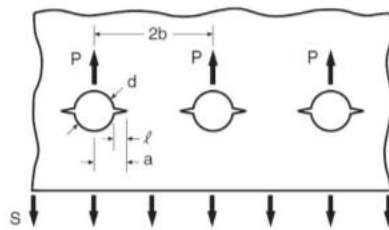


Figure 8 crack emanating from hole

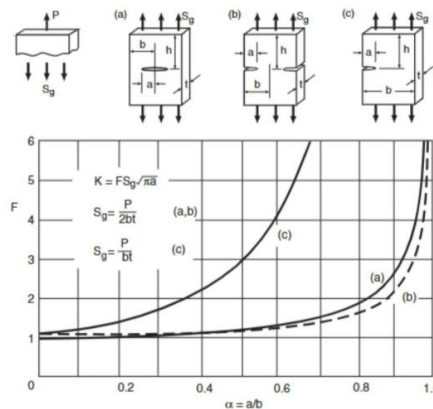


Figure 9 a/b vs F curves for different crack types

Distance from notch centre to crack tip is the sum of crack length and radius. The crack length is 1.5mm and radius is 3.175mm , based on this effective crack length (a) is 4.675mm , the distance from flange edge to notch centre can be approximated as around half the width of flange and hence 35mm .

$$\frac{a}{b} = \frac{4.65}{35} = 0.134 < 0.4 \text{ (mode a)}$$

Since a/b is less than 0.4 we can use

Equation 30

$$K = S_g * \sqrt{\pi * ai}$$

According to figure 9 we can use mode a where our S_g is the tensile stress on bottom flange.

$$S_g = \sigma_y = 95.71 \text{ MPa}$$

$$K = S_g * \sqrt{\pi * a_i}$$

$$K = 95.7 * \sqrt{\pi * 0.004675} = 11.59 \text{ MPa}\sqrt{\text{m}}$$

based on above we can calculate for final crack size against brittle fracture and plastic collapse and further determine critical mode of failure and its margin of safety.

$$X_k = \frac{K_{Ic}}{K} = \frac{26}{11.59} = 2.24$$

Equation 31

$$a_c = \frac{1}{\pi} * \left(\frac{K_{Ic}}{S_g} \right)^2$$

$a_c = 23.499 \text{ mm}$ (brittle fracture)

For plastic collapse, load to cause yielding,

Equation 32

$$P_0 = \sigma_0 * (b - a) * t$$

Hence load to cause yielding P_0 ,

$$P_0 = 441 * (70 - (2 * 1.5 + 6.35)) * 17 = 454,693 \text{ N}$$

$$P = \sigma_y * b * t$$

$$P = 114,109 \text{ N}$$

$$X_p = \frac{P_0}{P} = 4$$

Equation 33

$$a_p = b * \left(1 - \frac{S_g}{\sigma_0} \right)$$

Hence the margin of safety against brittle fracture (X_k) and plastic collapse (X_p) are 2.24 and 4 respectively for cracks emanating from hole on bottom flange. Hence critical mode of failure would be brittle fracture with a margin of safety of 2.24

Table 8 Fracture analysis for case 1

ai	0.004675	b
σ	95.71509788	
R	0.003175	
a/b	0.133571429	
F	1	
K	11.59674425	MPa√m
K _{Ic}	26	MPa√m
S ₀	441	
X _k	2.242008571	brittle fracture
a _p	54.80712732	mm
a _c	23.49936637	mm
P ₀	454693.05	N
P	113900.9665	N
X _p	4	

In case 2 that is centre crack in web we will start by determining the ratio of a and b with a being 1.5mm and b being 113mm (half the height of web) hence a/b is less than 0.4 hence using mode an again we will use equation 29, we use S_g as 83.19 which is the stress found at web flange junction.

$$K = S_g * \sqrt{\pi * a_i} = 83.19 * \sqrt{\pi * 0.0015} = 5.7 \text{ mpa}\sqrt{\text{m}}$$

based on above we can calculate for final crack size against brittle fracture and plastic collapse and further determine critical mode of failure and its margin of safety.

From equation 30,

$$ac = \frac{1}{\pi} * \left(\frac{K_{IC}}{S_g} \right)^2 = \frac{1}{\pi} * \left(\frac{26}{83.19} \right)^2 = 31 \text{ mm}$$

$$X_k = \frac{K_{IC}}{K} = \frac{26}{5.7} = 4.56$$

For plastic collapse, load to cause yielding,

$$P_0 = \sigma_0 * (b - a) * t$$

Hence load to cause yielding P_0 ,

$$P_0 = 441 * (226 - 1.5 * 2) * 17 = 226188.9 \text{ N}$$

$$P = \sigma_y * b * t$$

$$P = 43242.16 \text{ N}$$

$$X_p = \frac{P_0}{P} = 5.23$$

From equation 32,

$$ap = b * \left(1 - \frac{S_g}{\sigma_0} \right) = 226 * \left(1 - \frac{83.19}{441} \right) = 183 \text{ mm}$$

Hence the critical margin of safety for web for tensile loads is 4.56 against brittle fracture. For shear loading (mod II) the margin of safety is considerably high.

center crack in web		
a	0.0015	m
b	0.113	m
a/b	0.013274336	<0.4
F	1	
S_nominal	83.19	mPa
K	5.709283093	mpa√m
Xk	4.553986828	
ac	31.10819404	mm
ap	183.367483	mm
P0	226188.9	N
P	43242.162	N
Xp	5.230749101	

Figure 10 Fracture analysis for case 2

Finally edge cracks in flanges, in this we again can safely neglect the top flange due to compression, and focus on tensile stresses faced on the bottom.

$$\frac{a}{b} \ll 0.4$$

$F = 1.12$ (mode c)

Hence from equation 27,28,29,30,31

$$\begin{aligned}
 K &= F * \sigma * \sqrt{\pi * ai} \\
 K &= 1.12 * 95.71 * \sqrt{3.14 * 0.0015} \\
 K &= 7.35 \text{ mPa}\sqrt{\text{m}} \\
 K_{IC} &= 26 \text{ mPa}\sqrt{\text{m}} \\
 X_k &= \frac{K_{IC}}{K} = 3.53 \\
 ac &= \frac{1}{\pi} * \left(\frac{K_{IC}}{F * S_g} \right)^2 = \frac{1}{\pi} * \left(\frac{26}{1.12 * 95.71} \right)^2 = 18.72 \text{ mm} \\
 P_0 &= 441 * (70 - 1.5) * 17 = 513544 \text{ N} \\
 P &= \sigma_y * b * t \\
 P &= 114,109 \text{ N} \\
 X_p &= \frac{P_0}{P} = 4.5 \\
 ap &= b * \left(1 - \frac{S_g}{\sigma_0} \right) = 70 * \left(1 - \frac{95.7}{441} \right) = 54.8 \text{ mm}
 \end{aligned}$$

hence the critical margin of safety is 3.53 against brittle fracture.

edge crack on flange		
a	0.0015	mm
b	0.07	mm
a/b	0.021428571	
F	1.12	
ac	18.733551	mm
ap	54.80712732	mm
K	7.357138368	mpa√m
P0	513544.5	N
P	113900.9665	
Xk	3.533982739	
Xp	4.508693086	

Figure 11 Fracture Analysis of case 3

4.4 Results

It can be observed that the lift force acting on the wing with consideration to rib 9 is around 17329N which is reasonable in consideration to application of a safety factor of 1.5 (FAR 23, ASTM). The applied Kutta-Joukowski theorem helps us relate the circulation on wing to lift generated and incorporates the importance of vortices in generating lift. Furthermore, the application of elliptical lift distribution gives a nearly accurate distribution of lift similar to Prandtl's lifting line theory and software like XFLR. This lift force provides us with reasonable direct stresses of 95.89 and 83.4 mPa in compression on top and tension on the bottom flange which is significant with the observations made in flights that wings tend to deflect upwards in cruise due to compression and tension forces. (German Aerospace Center (DLR) 2020). These produce Margins of safety of 3.8 (point A) 3.5(B) and 10 (web) as observed from figure 5, these indicate that for a safety factor of 1.5 applied on the wing, which means for load 1.5 times the weight of the aircraft we observe a critical margin of safety of 3.5. Meanwhile for a safety

factor of 1, it would be observed to be around 6 which further strengthen our findings. (look at appendix to evaluate)

The notched stresses on flanges display elastic behaviour which explains and helps us understand that the spar flanges act within tolerable and safe limits. Meanwhile the analysis for shear web shows plastic deformation which may be due to consideration of shear strength as the deciding factor against nominal strength another reason can be the low thickness.

The fracture analysis results indicate that for the three primary crack scenarios on the spar—cracks emanating from holes on the flanges, centre cracks on the web, and edge cracks on the flanges—the spar has sufficient safety margins against brittle fracture and plastic collapse. In Case 1, cracks emanating from a hole on the bottom flange under tensile stress exhibit a critical margin of safety of 2.24 against brittle fracture, identifying brittle fracture as the primary concern for this scenario. For Case 2, the centre crack on the web demonstrates a higher margin of safety, with a value of 4.56, which signifies that the web is more robust against brittle fracture. For Case 3, edge cracks on the bottom flange show a margin of safety of 3.53, affirming sufficient resistance to brittle fracture.

The differences in margins of safety across the three cases—cracks emanating from holes on the flanges, centre cracks on the web, and edge cracks on the flanges—arise due to variations in stress concentrations and crack-tip stress intensity factors associated with each crack type and location.

Cracks emanating from holes on the flange (Case 1) have a lower margin of safety (2.24) compared to edge cracks (3.53) or centre cracks on the web (4.56), making this the most critical failure scenario. This is because holes introduce high stress concentrations due to the abrupt change in geometry, which intensifies the tensile stresses around the crack tip. As a result, even a small crack emanating from a hole can experience higher local stresses, increasing the likelihood of crack propagation and brittle fracture.

In contrast, edge cracks on the flange (Case 3) experience slightly lower stress concentrations, as the absence of an interior hole allows for a smoother stress distribution along the flange edge. Therefore, the stress intensity factor K_I and the resulting margin of safety are comparatively higher. Finally, the centre crack on the web (Case 2) has the highest margin of safety (4.56), as the larger surface area of the web can better distribute the tensile stresses, and the crack is more centrally located away from critical edges or geometric discontinuities, further reducing stress intensity at the crack tip.

In summary, cracks emanating from holes are more critical because holes act as strong stress concentrators, making the material around them more susceptible to fracture under tensile loading. This leads to a lower margin of safety for Case 1, emphasizing the need for careful design and reinforcement around holes in high-stress areas.

5 Dynamic Analysis

5.1 Introduction

Dynamic analysis in aircraft design involves assessing how structures respond to various movements, including vibrations and dynamic loading. This type of analysis goes beyond static calculations by evaluating the effects of cyclic and transient loads, which are crucial in understanding fatigue and structural integrity. For the Pilatus PC-9's wing main spar, dynamic analysis encompasses two essential fatigue assessment methods: safe-life and damage tolerance (also known as fail-safe design).

Both approaches are necessary, as each provides unique insights and safeguards that are critical to aircraft reliability, longevity, and safety under real-world operating conditions.

Safe-Life Analysis:

Safe-life is a preventive approach focused on predicting the number of load cycles the main spar can endure before fatigue cracks become a serious risk. This method is essential because, during flights, cyclic loads exert repeated stresses on the main spar, potentially initiating cracks that can grow undetected during routine inspections. Safe-life analysis calculates the maximum operational life of the component, providing a predetermined number of flight hours or load cycles before the part is retired from service. By using safe-life, engineers ensure the component can remain crack-free throughout its operational period, minimizing the risk of unexpected fatigue failure that could compromise structural integrity (Hoole, Sartor, & Cooper, 2016).

However, while the safe-life approach is effective in defining the part's maximum safe usage period, it does not account for scenarios where a crack has already developed. Without a complementary damage tolerance approach, unforeseen cracks could go undetected until they reach a critical size, leading to catastrophic failure.

Thus, safe-life is invaluable for standard operational limits but lacks the contingency planning required to address damage that could emerge between scheduled maintenance checks.

Damage Tolerance (Fail-Safe) Analysis:

Damage tolerance, or fail-safe design, is a complementary approach that considers the possibility of cracks or defects being present, especially around critical areas like fastener holes or stress concentrators. This method acknowledges that even under a safe-life framework, undetected damage or sudden crack growth may still occur due to manufacturing imperfections, accidental impacts, or unexpected stresses. Damage tolerance design thus aims to ensure that the main spar can sustain some level of damage without catastrophic failure, providing time for the crack to be detected and repaired during maintenance. This approach includes strategies like crack propagation analysis, fracture toughness testing, and the strategic placement of redundant load paths to prevent complete failure if part of the structure is compromised.

In scenarios where the main spar might experience accidental damage or higher-than-expected stress, damage tolerance provides a secondary safeguard. By designing the spar with sufficient redundancy and fracture resistance, fail-safe principles allow the structure to continue functioning, even with a crack, until corrective maintenance is possible.

This added resilience is especially valuable in cases of unforeseen loading conditions, where relying solely on a safe-life approach could increase risk.

Importance of Combining Safe-Life and Damage Tolerance:

Both safe-life and damage tolerance approaches are essential for a holistic fatigue assessment. While safe-life provides a foundation for crack-free operation within specified limits, damage tolerance ensures continued safety in cases where cracks or damage occur unexpectedly. Relying solely on safe-life could lead to overly conservative designs with frequent part replacements, increasing maintenance costs and downtime. On the other hand, depending only on damage tolerance would require exhaustive inspections and might not prevent initial crack formation.

Combining these methods provides a balanced strategy: safe-life ensures routine reliability by setting operational limits, while damage tolerance offers resilience against unexpected damage. Together, they enhance the main spar's safety and longevity, ensuring that the PC-9's wing structure can meet stringent performance and safety requirements, even under varied flight conditions and operational stressors. Has context menu

5.2 Safe life approach:

Although safe-life is a critical part of fatigue analysis, there are significant uncertainties since fatigue is a complex and probabilistic phenomenon. This causes limitations of predicting loads the main spar experiences during service and this due to the number of assumptions that are made to conduct the safe-life design process. In this scenario, the following assumptions made are:

- Fatigue damage model used. This can have many effects on the main spar's lifespan as it there are impacts on predicting lifespan as it provides a picture of how the material and the main spar will degrade overtime due to cyclic loading. On top of this, the calculation of margin of safety is another factor that impacts the model used.
- The material of the main spar, Al-2124-T851 will maintain its property throughout its service life.
- The environment conditions remain intact during the flight and will not cause any changes in the properties of the material.
- The loading and operational conditions such as pressure, temperature and load variations will remain constant meaning no extreme, or any unexpected loads will occur during the flight.
- Potential failure modes are identified to ensure the main spar has appropriate safety factors

Table 9Material Properties:

A	b = B
833.33	-0.079

The fatigue properties of Al-2124-T851 were chosen based on the experimental procedure carried out from the journal (Li Xue, Yin Zhimin, Nie Bo, Zhong Li, Pan Qinglin and Jiang Feng, 2007)

Table 10 PC-9 Loading Spectrum

Loading Condition	Normalised Stress Level	Cycles
I	0.5 to 1.0	1200
II	1.0 to 1.75	560
III	1.75 to 2.5	70
IV	2.5 to 3.5	25
V	3.5 to 5.0	30
VI	5.0 to 6.0	5
VII	6.0 to 7.0	25
VIII	1.0 to 4.5	6
IX	0 to 5.0	8
X	-3.5 to 1.5	25

Loading Case	LLR	HLR	Min Stress	Max Stress	σ_a	σ_m	σ_{ar}	Nf	Cycles	N/Nf
I	0.5	1	47.85	95.7	23.925	71.775	47.85	5.10551E+15	1200	2.3504E-13
II	1	1.75	95.7	167.475	35.8875	131.5875	77.52586	1.13585E+13	560	4.93022E-11
III	1.75	2.5	167.475	239.25	35.8875	203.3625	92.66113	1.18831E+12	70	5.89072E-11
IV	2.5	3.5	239.25	334.95	47.85	287.1	126.5992	22872684922	25	1.09301E-09
V	3.5	5	334.95	478.5	71.775	406.725	185.3223	183832949.8	30	1.63192E-07
VI	5	6	478.5	574.2	47.85	526.35	165.7573	754706114.9	5	6.6251E-09
VII	6	7	574.2	669.9	47.85	622.05	179.0383	284488038.2	25	8.78772E-08
VIII	1	4.5	95.7	430.65	167.475	263.175	268.5575	1679038.796	6	3.57347E-06
IX	0	5	0	478.5	239.25	239.25	338.3506	90170.44412	8	8.87209E-05
X	-3.5	1.5	-334.95	143.55	239.25	-95.7	185.3223	183832949.8	25	1.35993E-07
										9.26892E-05

Number of flights = 10788 to service. This is a safe number.

Computing the safe life according to the given loading spectrum on excel, the main spar has a lifespan of 10,000 cycles based on the max stress calculated from unnotched analysis of 95.7 Mpa in which inspection is required to replace the main spar to prevent any risk of failure. There could be many factors that can affect the result, and main factor is the assumptions made for the loading spectrum process. In this scenario, it is assumed that the material's properties will remain constant during the flight, the loading condition and margin of safety also has a significant impact on the safe-life result. In addition to this, the material properties are derived from the journal is specifically based on an experimental setup which we're trying to aim for.

5.3 Damage tolerant method

Ensuring structural safety in event of cracks or flaws is the major goal of the Damage Tolerance Methodology for deficiencies in the main spar. This method entails estimating the number of flight cycles required for cracks to reach a critical size that may cause failure and measuring the growth of fractures over time.

First, cracks are found by routine inspections, which are frequently conducted with non-destructive testing (NDT) techniques. Fracture mechanics is used to model the growth of a crack under operating stresses once it is discovered. Two potential failure mechanisms are identified by the analysis: plastic collapse, in which the material deforms before failure, and brittle fracture, which happens abruptly when a crack grows larger than a critical size.

Engineers can determine how many flights cycles the crack will withstand before it becomes harmful by projecting the rate at which the crack will grow. This makes it possible to set up

suitable inspection intervals to track the development of the fracture and take action before it gets to a critical size.

The process makes it possible to make well-informed conclusions regarding the structural integrity of the main spar and assists in deciding if repairs, replacements, or continuous operation are required. It is essential to preserving the spar's longevity and safety throughout use by averting unanticipated failures brought on by flaws.

Here we observe three cases:

- Crack emanating from hole on flange
- Centre crack in web
- Edge crack on flange

All the cases are based on tensile load (bottom half of the spar)

Procedure for calculating fatigue life for variable load spectrum

1. Calculate the equivalent zero-to-tension stress value corrected for the R effect for each block:

$$\overline{\Delta S} = S_{max}(1 - R)^{\gamma}$$

Equation 34

2. Calculate the equivalent zero-to-tension stress level for the whole spectrum

$$\Delta S_q = \left[\frac{\sum_{j=1}^{N_B} (\overline{\Delta S}_j)^m}{N_B} \right]^{1/m} = \left[\frac{\sum_{j=1}^k (N_j (\overline{\Delta S}_j)^m)}{N_B} \right]^{1/m}$$

Equation 35

3. Use Paris equation to calculate number of cycles:
 1. a_f needs to correspond to the most severe force in the spectrum (brittle fracture & plastic collapse)
 2. Check the validity of F.
4. Calculate the number of repetitions:

$$B_{if} = \frac{N_{if}}{N_B} \quad (N_B = \sum_{j=1}^k N_j)$$

Equation 36

For case 1: Crack emanating from hole on flange

ai	0.004675
σ	95.71509788 mpa

Loading Case	LLR	HLR	Min Stress	Max Stress	Cycles	R	Delta_s	Nj(deltaS)^m
I	0.5	1	47.85	101	1200	0.47376238	19.02805	6157787.8
II	1	1.75	95.7	167.475	560	0.57142857	18.50165	2649098
III	1.75	2.5	167.475	239.25	70	0.7	10.45604	63279.486
IV	2.5	3.5	239.25	334.95	25	0.71428571	12.89446	41505.811
V	3.5	5	334.95	478.5	30	0.7	20.91208	202429.19
VI	5	6	478.5	574.2	5	0.83333333	5.443396	680.75811
VII	6	7	574.2	669.9	25	0.85714286	4.253585	1664.676
VIII	1	4.5	95.7	430.65	6	0.22222222	224.052	39279220
IX	0	5	0	478.5	8	0	478.5	472876377
X	-3.5	1.5	-334.95	143.55	25	-2.3333333	143.55	45003810

Number of cycles $N_B = 1954$

Fatigue cycles $N_{If} = 9.37E+07$

Number of cycles to repair $B_{if} = \frac{N_{if}}{N_b} = 47900$

For case 2: center crack in web (under shear)

ai	0.0015
σ	209 mpa

Loading Case	LLR	HLR	Min Stress	Max Stress	Cycles	R	Delta_s	Nj(deltaS)^m
I	0.5	1	104.5	209	1200	0.5	34.472144	34501605.77
II	1	1.75	209	365.75	560	0.571429	40.405891	25519821.76
III	1.75	2.5	365.75	522.5	70	0.7	22.835028	609596.6272
IV	2.5	3.5	522.5	731.5	25	0.714286	28.160309	399842.0934
V	3.5	5	731.5	1045	30	0.7	45.670057	1950081.479
VI	5	6	1045	1254	5	0.833333	11.887877	6558.015402
VII	6	7	1254	1463	25	0.857143	9.2894376	16036.49055
VIII	1	4.5	209	940.5	6	0.222222	489.30897	378392457.1
IX	0	5	0	1045	8	0	1045	4555407467
X	-3.5	1.5	-731.5	313.5	25	-2.33333	313.5	433539716.9

Number of cycles $N_B = 1954$

Fatigue cycles $N_{If} = 3.02E+07$

Number of cycles to repair $B_{if} = \frac{N_{if}}{N_b} = 15,500$

For case 3: edge crack on flange:

ai	0.0015
σ	95.7 mpa

Loading Case	LLR	HLR	Min Stress	Max Stress	Cycles	R	Delta_s	Nj(deltaS)^m
I	0.5	1	41.5	83	1200	0.5	13.689895	2369962.895
II	1	1.75	83	145.25	560	0.571429	16.046359	1752991.761
III	1.75	2.5	145.25	207.5	70	0.7	9.0684563	41874.03326
IV	2.5	3.5	207.5	290.5	25	0.714286	11.183281	27465.70498
V	3.5	5	290.5	415	30	0.7	18.136913	133953.7869
VI	5	6	415	498	5	0.833333	4.7210227	450.4791247
VII	6	7	498	581	25	0.857143	3.6891068	1101.568658
VIII	1	4.5	83	373.5	6	0.222222	194.31887	25992299.87
IX	0	5	0	415	8	0	415	312917223
X	-3.5	1.5	-290.5	124.5	25	-2.33333	124.5	29780441.22

Number of cycles $N_B = 1954$

Fatigue cycles $N_{If} = 2.08E+06$

Number of cycles to repair $B_{if} = \frac{N_{if}}{N_b} = 1060$

5.4 Results

For crack emanating from flanges, we calculate a damage tolerant life of 47900 flights to inspection, while for centre crack in web it yields a result of 1060 flights this can also be seen from notched analysis of web where it yields plastic stresses on unloading which indicates that web needs to be inspected carefully. The assumptions used also lead to discrepancies in the finding as we assume web is active only in shear. Over all the web should have higher life than 1060 which can be determined accurately after in depth analysis. The third case with edge crack showcases a life of 15500 flights.

6 Experimental Analysis

6.1 INTRODUCTION:

When done correctly, experimental analysis can be the most appropriate way of determining accurate data. While computer-based models like as finite element analysis (FEA) have made tremendous progress in predicting the behavior of aircraft structures, real-world testing remains critical. As such, an experimental investigation was conducted, and the results will be used to verify other portions of the report.

Experimental testing gives insight into how the wing responds to operating loads in a simulated environment. This will help ensure that theoretical models relate to real world applications and aids fulfil stringent safety requirements. This experiment aims to better understand the physical properties of a PC9 aircraft by adding force to the wing simulating the pressures while in flight. The strain was measured using a delta rosette gauge which is one of the most accurate methods for determining the structural reaction of an aircraft's wing. Strain gauges capture localized data on how materials deform under stress by measuring strain in crucial areas of the wing.

The goal of this experiment is to apply controlled loads to a PC9 wing unit and determine the resulting strain with strain gauges. The strain data will then be analyzed and simplified for pertinent data, such as stress distribution across the wing. Stress values will be derived, which enables an evaluation of the wing's structural reaction. By attaching strain gauges to specific locations on the wing we can measure the strain over the displacement acting on the structure during flight condition. These readings enable us to determine the related principal stresses and strains, torsion and von mises stress, providing information about the wing's behavior under varied loading circumstances. The processed results will be evaluated to determine their reliability and validity, emphasizing any differences between experimental data and theoretical predictions.

6.2 EXPERIMENT SETUP

The experimental investigation is intended to analyse the structural behaviour of the PC-9 aircraft wing under simulated aerodynamic stress conditions, with rectangular strain gauge rosettes used to collect detailed strain data at significant locations on the wing structure.

Wing Mounting and Clamping:

The wing was mounted at the root and extended out freely. It was then clamped from the dihedral (rib 11) with a wood mould of the aerofoil that wrapped the cord of the wing. This enabled the replication of bending and shear pressures while maintaining a reusable and safe setup.

Load mechanism:

Two actuators were connected to the clap at rib 11 to simulate aerodynamic bending loads on the wing. The actuator applied regulated loading to imitate lift forces, resulting in an upward

deflection in the wingtip. The data collecting system was synced with the load cells and strain gauges, recording strain values as loads increased to develop strain-load correlations at various places on the wing.



Figure 12 Wooden clamping on wing with two actuators below at rib 11

Displacement Measurement:

A measuring tape was attached to the wingtip to monitor vertical wing tip displacement during loading. This method has a degree of accuracy of one millimeter however human factors such as mismanagement of the measuring tape may lead to larger inaccuracy of results.



Figure 13 Measurement tape for displacement

As well as the wing tip displacement, the displacement at the position of load (rib 11) was also taken. This was measured using a mechanical devise that would tension a wire and use the force to calculate the displacement. This method is more accurate than the measuring tape.

Strain Gauge Integration:

Five rectangular strain gauge rosettes were strategically placed on the wing to measure strain in multiple directions. Each rosette was made up of three gauges that measured strain in different orientations (usually 0° , 45° , and 90°) relative to the structure. The position of the gauges can be seen below.

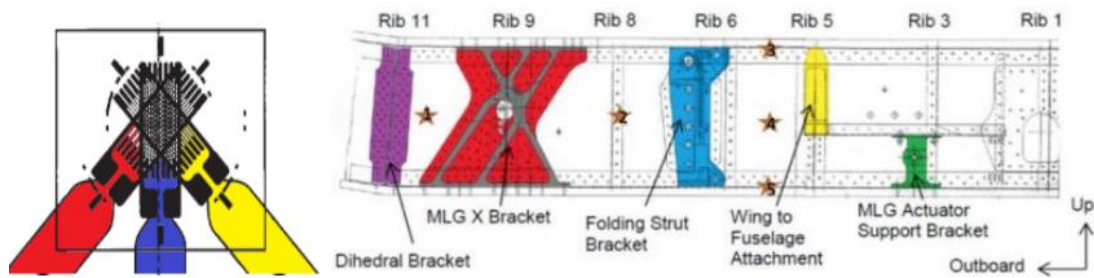


Figure 14: position of sensors



Figure 15 strain gauge sensor

The gauges were placed at the top spar cap, between ribs 5 and 11 (main bending load region). This can be seen in figure 14 represented as stars



Figure 16 Rib-6

- Bottom spar cap (opposite side for measuring bending strain).
- Shear web in the main spar (to catch shear strain during loading).
- Wing skin on rib 8 (to capture skin deformation caused by wing flexion).



Figure 17 Rib-9 (MLG-X)

- At rib 11, toward the beginning of the dihedral portion (to determine strain near the clamped region).

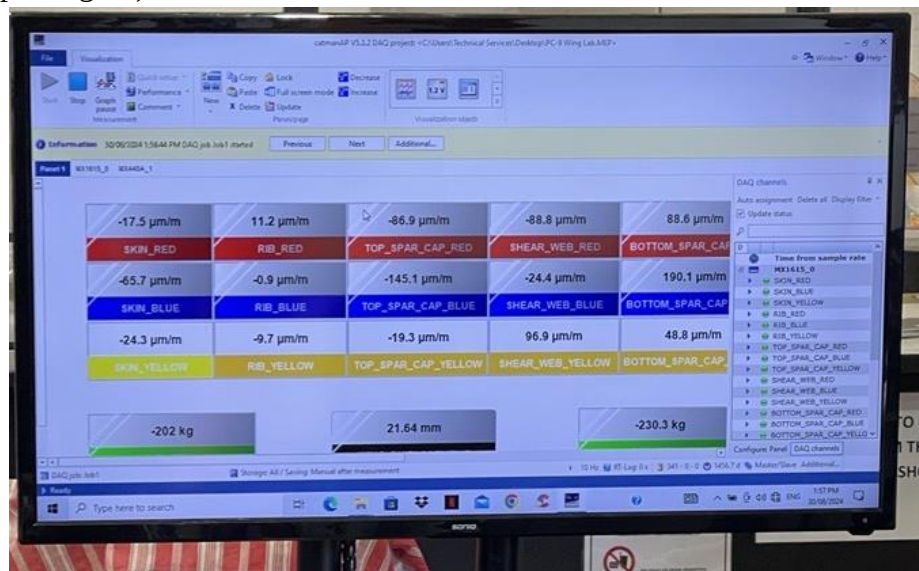


Figure 18 Displacement, load, stress and strain values will be visible on screen as in above picture

The data collected from the gauges was put into an excel sheet divided into 3 sections, Red, Blue and Yellow. These three measurements are then used to calculate the strain in x, y, and xy directions with the equations below. The units measured were in $\mu\text{m/m}$ and were then converted to mm/mm later in this report.

Equation 37: strain calculations

ϵ_x	$=\epsilon_{\text{red}}$
ϵ_y	$=\epsilon_{\text{yellow}}$
γ_{xy}	$= 2\epsilon_{\text{blue}} - \epsilon_{\text{red}} - \epsilon_{\text{yellow}}$

When calculating these values, the average strain, displacement and force was taken for each load case.

6.3 PROCEDURE:

Calibration of the system was done before the experiment to guarantee precise findings. The strain gauges were set to zero to record only new strain, and the hydraulic actuator was tested to apply smooth loads. After the data is stabilized, the wing was then subjected to progressively increasing loads, and for each step, the strain and displacement were recorded. At strategic locations on the wing, data from displacement sensors, load cells, and strain gauges were continuously gathered. After calculating the direct strains and stress values, the values can be compared to the Finite Element Analysis.

Data Collection: The strain data obtained from the rosettes provided information on the wing's structural response. The strain measurements were used to calculate primary and shear strains as well as principal stress and von mises stress which are critical for understanding the stress distribution and potential weak points in the wing structures recorded by the computer in excel sheet.

6.4 RESULTS:

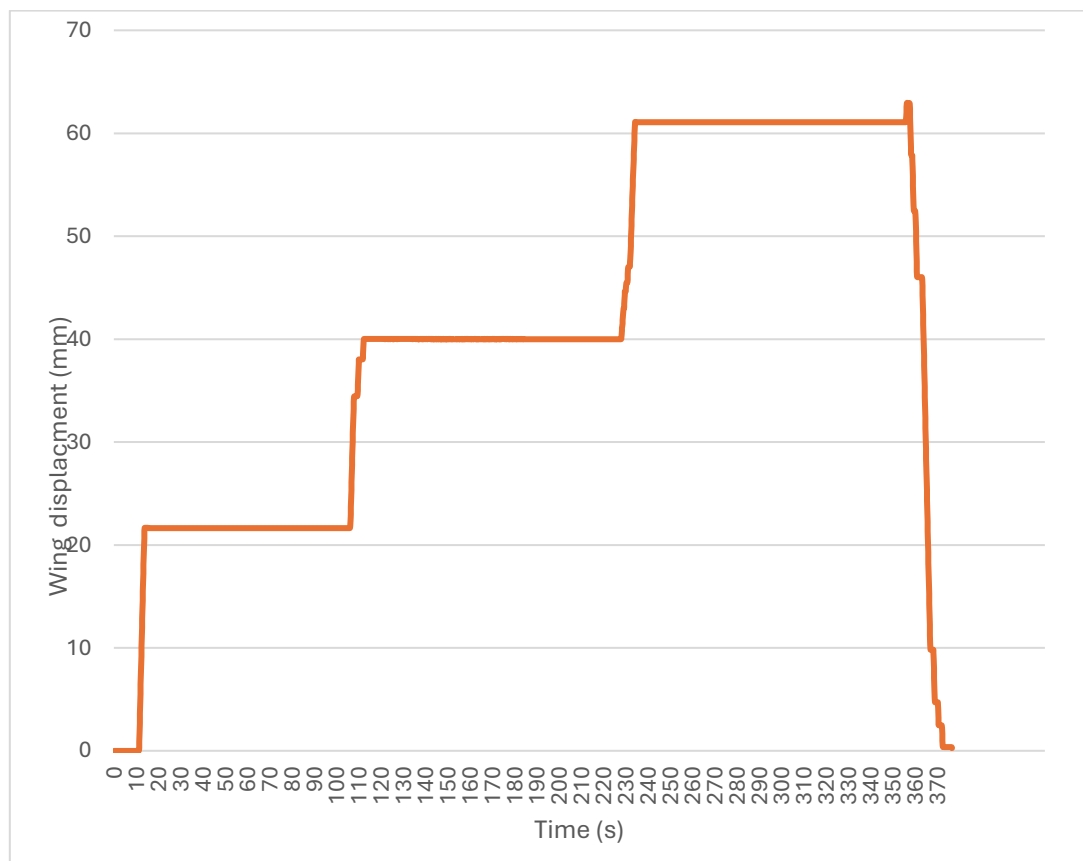


Figure 19: wing displacement at load (mm) over time(s)

The wing displacement against time graph shows the wing displacement over time is divided into three separate stages which are the three loading conditions. The displacement reaches about 21.62 mm during the first stage, which lasts for about 100 seconds. The displacement increases to about 40 mm during the second stage, which lasts for about 100 seconds. The

displacement finally peaks at about 61.06 mm in the third stage, which occurs for 120 seconds. Following this peak, the displacement quickly drops to zero at 370 seconds, which indicates the experiment's unloading phase.

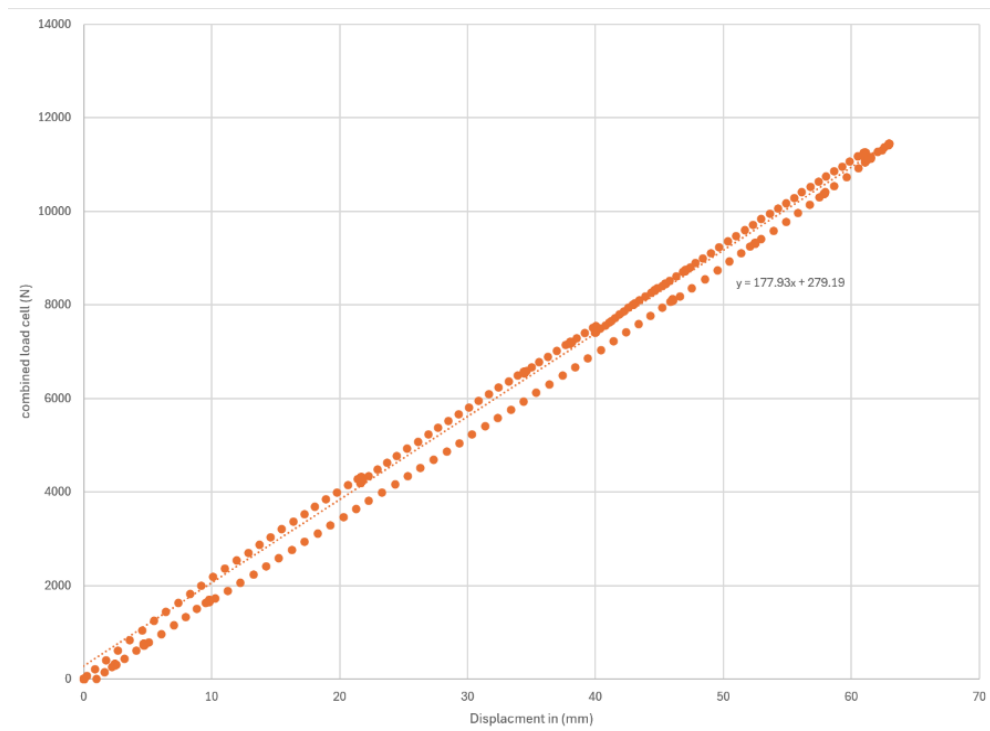


Figure 20: force (N) over displacement (mm)

The plot above shows there is a linear relationship between combined load cell and displacement. This means that the displacement is directionally proportional to the combined load cell. The load was calculated by adding both load cells and multiplying the resultant by 9.8 to convert the force from kg to N.

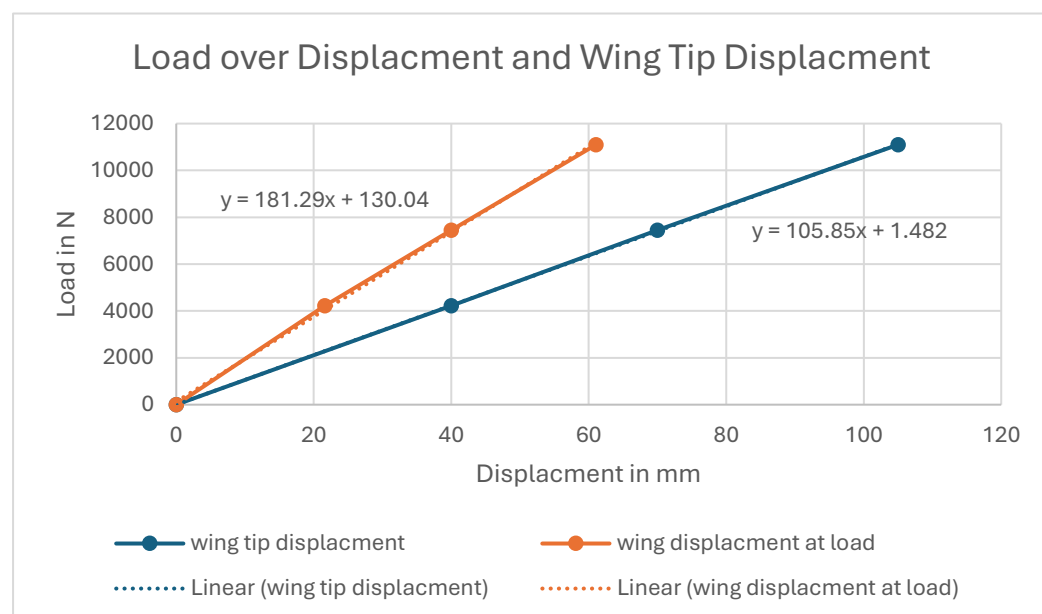


Figure 21: comparison of wing tip displacement and displacement at load overload

According to the equation's $Y=177.93X+279.19$, every millimetre of displacement results in an approximate increase in load of 177.93 N. An initial preload or residual stress in the wing structure could potentially be shown by the y-intercept at 279.19 N. Up to this displacement range, a predictable elastic response in the wing is implied by the consistency of the data points surrounding the line. This linearity, which is in good alignment with predicted elastic behavior, verifies that the wing's material and structural arrangement sustain a steady deformation pattern under increasing loading.

Table 11: displacement and load results for each load case

	Displacement at load (mm)	Displacement at wing tip (mm)	Load (N)
Load case 1	21.62	40	4220.44
Load case 2	40.06	70	7444.04
Load case 3	61.06	105	11100.2
Equation of the line = $Y=177.93X+279.19$			

From our calculated strains, we can calculate the principal strains using the equation:

Equation 38:

$$\varepsilon_{1,2} = \frac{\varepsilon_x - \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

Where: $\varepsilon_1 > \varepsilon_2$

γ_{max} is then calculated using the equation:

Equation 39:

$$\gamma_{max} = 2 \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

The angle in radians can be calculated using:

Equation 40:

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \right)$$

The principal stress can then be calculated with the assumptions of hooks law using the equations:

Equation 41:

$$\sigma_1 = \left(\frac{E}{1 - \nu^2} \right) \times (\varepsilon_1 + \nu \varepsilon_2)$$

Equation 42:

$$\sigma_2 = \left(\frac{E}{1 - \nu^2} \right) \times (\varepsilon_2 + \nu \varepsilon_1)$$

Where: $\sigma_1 > \sigma_2$

Finally the max torsion can be calculated using:

Equation 43:

$$\tau_{max} = abs\left(\frac{\sigma_1 - \sigma_2}{2}\right)$$

We then use these equations on each load case resulting in the following tables:

Load Case 1

Load Case 1: 21.62 mm						
	skin	rib	top spar	shear web	bottom spar	
ϵ_x	-0.0000180	0.0000108	-0.0000868	-0.0000887	0.0000880	mm/mm
ϵ_y	-0.0000245	-0.0000100	-0.0000245	0.0000963	0.0000485	mm/mm
γ_{xy}	-0.0000892	-0.0000031	-0.0001773	-0.0000578	0.0002424	mm/mm
ϵ_1	0.0000235	0.0000109	0.0000383	0.0001008	0.0001911	mm/mm
ϵ_2	-0.0000659	-0.0000101	-0.0001496	-0.0000931	-0.0000546	mm/mm
γ_{max}	0.0000894	0.0000210	0.0001879	0.0001939	0.0002456	rad
θ	-0.7487646	-0.0737117	0.6165249	0.1513395	0.7045665	rad
σ_1	0.1407969	0.6232051	-0.9086523	5.7434167	14.1956751	Mpa
σ_2	-4.7729821	-0.5319835	-11.2370477	-4.9135297	0.6964772	Mpa
σ_{eq}	4.8449151	1.0014619	10.8113977	9.2385095	13.8605667	Mpa
τ	-2.4568895	-0.5775943	-5.1641977	-5.3284732	-6.7495990	Mpa

Load Case 2

Load Case 2: 40 mm						
	skin	rib	top spar	shear web	bottom spar	
ϵ_x	-0.0000283	0.0000225	-0.0001720	-0.0001701	0.0001437	mm/mm
ϵ_y	-0.0000365	-0.0000157	-0.0000122	0.0001849	0.0000977	mm/mm
γ_{xy}	-0.0001558	-0.0000063	-0.0003276	-0.0000816	0.0004252	mm/mm
ϵ_1	0.0000456	0.0000227	0.0000901	0.0001896	0.0003346	mm/mm
ϵ_2	-0.0001104	-0.0000159	-0.0002744	-0.0001747	-0.0000931	mm/mm
γ_{max}	0.0001560	0.0000386	0.0003645	0.0003643	0.0004277	rad
θ	-0.7591889	-0.0818827	0.5585271	0.1129589	0.7315214	rad
σ_1	0.7540633	1.4330103	-0.0346947	10.8225052	24.9248584	Mpa
σ_2	-7.8193486	-0.6910565	-20.0695410	-9.1985082	1.4193758	Mpa
σ_{eq}	8.2223542	1.8765311	20.0522162	17.3577094	24.2463492	Mpa
τ	-4.2867060	-1.0620334	-10.0174232	-10.0105067	-11.7527413	Mpa

Load case 3

Load Case 3: 61.06 mm						
	skin	rib	top spar	shear web	bottom spar	
ϵ_x	-0.0000397	0.0000294	-0.0002918	-0.0002682	0.0002027	mm/mm
ϵ_y	-0.0000578	-0.0000234	0.0000235	0.0002879	0.0001540	mm/mm
γ_{xy}	-0.0002347	-0.0000093	-0.0005364	-0.0001043	0.0006328	mm/mm
ϵ_1	0.0000690	0.0000298	0.0001769	0.0002928	0.0004957	mm/mm
ϵ_2	-0.0001665	-0.0000238	-0.0004453	-0.0002731	-0.0001390	mm/mm
γ_{max}	0.0002354	0.0000536	0.0006222	0.0005658	0.0006346	rad

θ	-0.7467557	-0.0874530	0.5196774	0.0926634	0.7470012	rad
σ_1	1.1506331	1.7994290	2.4614715	16.6241999	36.8997051	Mpa
σ_2	-11.7892160	-1.1484718	-31.7354994	-14.4746650	2.0186086	Mpa
σ_{eq}	12.4046215	2.5736211	33.0350843	26.9538434	35.9329508	Mpa
τ	6.4699246	1.4739504	17.0984855	15.5494325	17.4405483	Mpa

The following values were used for Young's modulus and poisons ratio:

$$E = 7.31E+04 \text{ MPa}$$

$$\nu = 0.33$$

DISCUSSION

Our experimental investigation's results show that the PC9 wing demonstrated clear, measured displacements under incremental loading circumstances, with the following data points noted:

- 21.62 mm corresponding to a load of 4220.44 N,
- 40 mm corresponding to a load of 7444.04 N,
- 61.06 mm corresponding to a load of 11100.21 N.

The computational Finite Element Analysis (FEA) model of the wing will be validated using these displacements, which are important empirical reference points. The dataset adds to a better knowledge of the wing's structural performance under load by including metrics such as Von Mises stress, shear stress, maximum stress, principal strain, and principal stress at important points on the wing structure.

These displacements provide crucial data to support the validity of the Finite Element Analysis (FEA) model, as they are corresponding to expected structural responses. At five crucial locations on the wing's structure, important mechanical data including principal stress, principal strain, maximum stress, maximum strain, shear stress, and Von Mises stress were recorded in addition to the observed displacements. The analysis shows that the wing's various parts react differently to the imposed loads.

The following analysis is for load case 3:

The provided data offers detailed insights into the stress and strain distributions across various structural elements of a plane wing, including the skin, rib, top spar, shear web, and bottom spar. Beginning with principal strains (ϵ_1 and ϵ_2), we observe that ϵ_1 is positive across most components, indicating tensile strains, while ϵ_2 is negative, showing compressive strains. Notably, the bottom spar has the highest tensile strain ($\epsilon_1 = 0.0004957 \text{ mm/mm}$), suggesting it undergoes the most significant elongation. Conversely, the rib exhibits minimal strain, indicating that it may be under lighter loading conditions compared to other parts.

In terms of maximum shear strain (γ_{max}), the bottom spar again leads with the highest value (0.0006346 rad), which implies it is subjected to considerable torsional loading. High shear strains are also seen in the top spar and shear web, aligning with their structural functions as they typically endure substantial shear forces. The rib, with the lowest shear strain, likely serves a supportive role rather than carrying heavy loads.

The orientation of the principal stresses (θ) varies significantly among components. The top and bottom spars exhibit larger values of θ , but with opposite signs, suggesting a distinct orientation of stress directions, possibly due to differing load distributions or material anisotropy. Meanwhile, the skin

shows a slight negative θ , oriented in opposition to the bottom spar, possibly indicating bending loads acting on the wing structure.

The principal stresses (σ_1 and σ_2) further highlight the varied loading conditions across components. The bottom spar experiences the highest tensile stress ($\sigma_1 = 36.8997$ MPa), while the skin and rib show lower values, suggesting they are less stressed overall. The top spar faces the highest compression ($\sigma_2 = -31.7355$ MPa), consistent with typical wing bending behaviour, where the upper spar is subjected to compressive forces. Both the shear web and bottom spar show high tensile and compressive stresses, indicative of combined bending and torsional loads.

Looking at the equivalent stress (σ_{eq}), which combines the stress states to predict yield potential, the bottom spar and shear web stand out with high values, marking them as critical areas where failure is more likely under the current load. The top spar also has a significant equivalent stress, reflecting its role in resisting bending forces. Meanwhile, the skin and rib have relatively low σ_{eq} values, indicating they are under less intense loading.

Finally, shear stress (τ) distribution reveals that the bottom and top spars bear the highest values, further reinforcing their role in resisting both bending and torsional effects. The rib, in contrast, has the lowest shear stress, aligning with its supportive function rather than direct load bearing in bending or torsion.

To ensure the structural integrity and safety of the PC9 aircraft, it is crucial to handle both tensile and shear stresses at different positions along the wing, as highlighted by the experimental data. Future design developments will be influenced by these insights, especially in regions that are vulnerable to significant stress and deformation.

POTENTIAL SOURCES OF ERROR:

The reliability and accuracy of the collected data must be ensured by closely evaluating potential causes of error, as with any experimental setup:

- **Measurement Accuracy:** The digitally recorded displacement values may vary somewhat but the displacement measured manually may vary significantly if the tape measure used warped or moved during the measuring. When compared to theoretical expectations, errors as small as millimetres can cause variances.
- **Actuator Calibration:** If the loading actuators' calibration was not precisely aligned, there may be differences between the intended and actual actuator performance. The consistency of the loading conditions throughout successive test iterations may have been impacted by minor variations between displacement and force application introduced by the manual setting and control of the two actuators.
- **Manual Control of Load Application:** Slight deviations in the recorded load-displacement relationship can be interpreted as non-linearities or variances in the displacement curves caused by human intervention in manually managing the load increments following each application.

The experimental procedures adhered to standardized best practices for the structural testing of aerospace components, and due diligence was taken to minimize the potential sources of error. The comparatively small magnitude of these errors indicates that they have a minimal impact on the final outcomes and have no discernible effect on the overall inferences made from the experimental data.

AUTHENTICITY OF EXPERIMENTAL DATA:

The experimental data is regarded as reliable and authentic despite the setup's inherent constraints and the potential sources of error that have been found. Under realistic loading conditions, the observed displacement patterns and the accompanying load levels are in alignment with the theoretical expectations for the PC9 wing. This substantial level of consistency lends credibility to the empirical results, which are an essential means of validating the FEA model that was built simultaneously. Although slight discrepancies between numerical and experimental data are to be expected because both physical and computational models are complicated, the alignment of the two increases the validity of the findings. Therefore, useful empirical evidence from the experimental inquiry has been obtained, which may be used to verify the computer estimations. These observations will provide a basis to make further design improvements that maximize structural safety and wing performance.

6. FEA MODELLING

INTRODUCTION:

A vital resource for developing and evaluating aircraft structures is finite element analysis (FEA). Because aircraft components are made of many different parts and are subjected to adverse environments, finite element analysis (FEA) enables to model and forecast how these structures will respond to different loads and environmental stresses such as stress distribution, deflections and failure points. This has a direct impact on operational longevity and fuel efficiency. Using finite element analysis (FEA), complex constructions can be dissected into smaller, more manageable components to allow for a complete examination of the distribution of forces, including shear, torsion, and bending, inside the material. This ability is essential for improving the design and reducing weight without compromising the aircraft's strength or integrity.

FEA will be used during this project to simulate the PC-9 aircraft wing. The primary objective is to verify the structural integrity of the aluminium 2124-T851 alloy wing by comparing the FEA results with experimental strain gauge data. Aluminium alloy 2124-T851 has a high strength-to-weight ratio, under increasing Design Ultimate Load (DUL) scenarios. All the stresses mentioned above will be simulated using the FEA model, which will offer information on stress distributions, probable failure sites, and general wing behaviour under load. We hope to evaluate the accuracy of our design and identify areas for improvement by contrasting the simulation with actual strain results.

The Numerical Methods

Simplifying the PC9-Wing Modelling Process Simplifications:

To minimize computational effort, a few small parts, such as stringers, tiny holes, and non-load-bearing features, were removed while keeping the primary structural elements, such as spars, ribs, and skins. With minimal influence on accuracy, this guarantees computational efficiency

Justification: The premise that minute geometric details have no effect on the overall stress distribution and have no relevance on the structure's primary load-carrying behavior serves as justification for these simplifications.

FE Model Details

Material: Al2024-T851,

Element Type: To accurately depict the intricate geometry of the wing, linear or quadratic tetrahedral elements were chosen.

Mesh: A sensitivity analysis of the mesh was performed (Table 1). To observe the effect on stress and displacement findings, different mesh sizes were used. Based on convergence data, a particular mesh size for several components was determined to be best (see below).

Loads and Boundary Conditions (BCs)

Limitations: The wing's root is represented as a stationary support, mimicking its attachment to the fuselage

Application of Load: In order to simulate the aerodynamic loads encountered during flight, a distributed pressure load is applied to the wing surface.

Limitations: The wing's root imposes constraints on all the degrees of freedom, leaving the other surfaces unrestricted in their ability to flex under the given force.

PARAMETER	Case1	Case2	Case3
Material	Al-2124-T851	Al-2124-T851	Al-2124-T851
Mean element size	0.03m	0.03m	0.03m
No of elements	143498	143498	143498
Nodes	148906	148906	148906
Load	104.35 N/mm	137 N/mm	182 N/mm

Mesh Sensitivity Analysis

A critical phase in the FE modelling process is the mesh sensitivity analysis, which makes sure that the simulation results accurately reflect the behavior of the structure under load and are not disproportionately impacted by the size of the mesh elements. A methodical mesh refinement analysis was carried out for this project in order to find the best possible balance between computing efficiency and accuracy. To make sure that the findings converged, the mesh was gradually refined while the impacts on important outputs, including von Mises stress, strain, and deflection, were monitored.

The following procedures were involved in the mesh sensitivity analysis:

Initial Mesh Generation:

Approximately 3.47m was the default element size when the first mesh was created. Tetrahedral solid elements were used to create this mesh in order to capture the intricate geometry of the PC9 wing.

Gradual Improvement:

The element size was gradually decreased (for example, from 3.47 m to 2.5m(minimum)) in order to refine the mesh in several stages.

To look for any notable variations in the stress, strain, and deflection findings, each refinement level was examined.

Convergence

When additional mesh size reduction resulted in changes to important outputs, including maximum stress and deflection, of less than 2%, convergence was considered to have been reached. By avoiding sensitivity to mesh size, this criterion guarantees that the results faithfully depict structural behavior.

Element Size(m)	Von mises Stress (Pa)	Deflection(m)	Simulation time(min)
3.47	630	1.29	5
3.0	603	1.27	7
2.7	587	1.26	9
2.5	579	0.64	11

Table 12 Mesh sensitive analysis

Remarks:

Convergence Achievement: The von Mises stress and deflection values exhibit less than a 2% change from the prior refinement level at a mesh size of 2.5 cm, suggesting convergence.

Computational Efficiency: The trade-off between accuracy and computational cost was highlighted when the element size was reduced from 3.47 cm to 2.5 cm, which resulted in an increase in simulation duration from 5 to 11 minutes.

Final Choice of Mesh

A mesh element size of 2.5 cm was chosen for the final model because it offered the best trade-off between computational efficiency and solution correctness. This mesh size is sufficiently fine-tuned to precisely capture deflection patterns and stress concentrations without requiring an excessive amount of calculation time. The final mesh can be seen in Fig

Dependability and Importance

The dependability of the FE model is increased by the mesh sensitivity study, which verifies that the selected element size of 2.5 cm yields mesh-independent findings. By confirming that predictions match actual structural behavior in the PC9 wing model, this validation step increases trust in the structural insights and design suggestions obtained from the research. In aircraft applications, where design efficiency and safety are top priorities, this thorough mesh selection procedure is crucial.

Validation of FE Model

The outcomes were compared to experimental data, such as load-displacement curves and von Mises stress distributions, in order to confirm the validity of the FE model.

CASE NUMBER	EXPERIMENTAL LOAD	FEA LOAD (N/mm)	EXPERIMENTAL DISPLACEMENT (mm)	FEA DISPLACEMENT
CASE 1	104.35	104.35	21	49.5
CASE 2	137	137	42	64.9
CASE 3	182	182	63	86.29

Table 13 FEA model validation

The FE model shows accurate load prediction across all cases, aligning with experimental loads (104.35 N/mm, 137 N/mm, and 182 N/mm), indicating reliable load calibration. However, it consistently overestimates displacement values compared to experimental results: in Case 1, the FEA displacement is 49.5 versus 21 experimentally; in Case 2, it's 64.9 versus 42; and in Case 3, it's 86.29 versus 63. A key factor contributing to this discrepancy is the thinner skin thickness (0.8 mm) used in the FEA model compared to a higher thickness in the experimental setup. Thicker skin in the experiment would increase structural stiffness and reduce displacement, explaining the model's overestimation. Other potential sources of variance include differences in material properties, boundary conditions, and model assumptions, which may also affect stiffness and displacement accuracy. Therefore, while the FE model effectively replicates load conditions, adjustments to parameters affecting stiffness, particularly skin thickness, are recommended for improved displacement accuracy. The pictures of the FE analyses are in the appendix

RESULTS

Results of stress, strain, and deflection from the FE analysis were compared for each of the three load instances. This is an overview:

PARAMTER	CASE1	CASE2	CASE3
Displacement (m)	0.49474	0.64954	0.86289
Von-mises stress (MPa)	579.5	760.81	1010.07
Shear stress (MPa)	12.008	15.765	20.94
Shear strain (mm/mm)	4.3694e-5	5.7366e-5	7.6209e-5
Principal stress (MPa)	317.56	416.93	553.87
Principal strain (mm/mm)	5.7851e-3	7.5952e-3	1.009e-2

Table 14 Three cases result for each parameter

Significant deformation under the design loads is demonstrated by the stress and deflection findings, with Case 3 exhibiting the highest von Mises stress. Structural safety was ensured as all cases stayed below the yield strength of Al2124-T851.

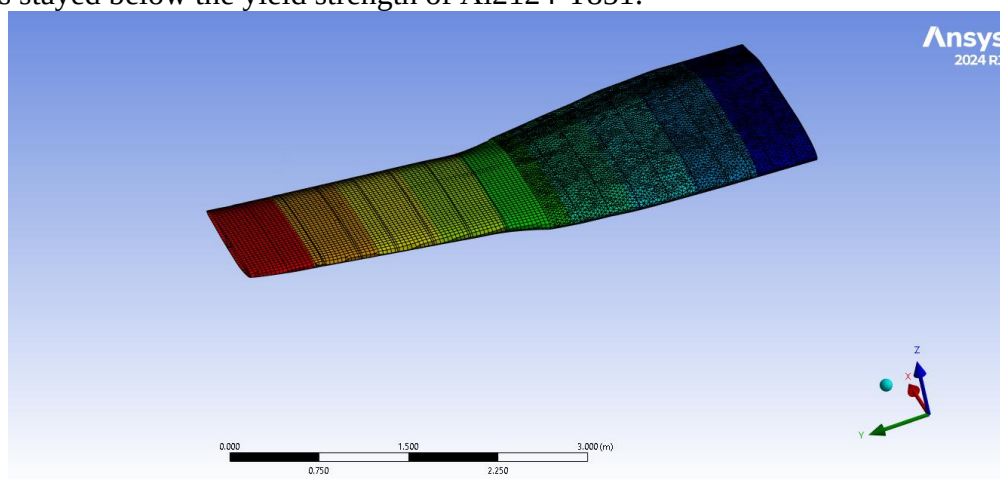


Figure 22 FE analysis for aluminium

More pictures of analysis can be found on [appendix](#)

Discussion

- The findings of the FE study offer a thorough comprehension of the PC9 wing's structural behavior under applied DUL conditions. Among the noteworthy discoveries are:
- The wing design performs linearly within the elastic range, as seen by the maximum stress increasing proportionately with the applied load. As the load increased, so did the deflection; in Case 3, the maximum deflection was 2.25 meters. This implies that while the wing can support heavy loads, more strengthening would be needed to maintain allowable deflection limits in harsh circumstances.
- The FE model's credibility for simulating structural reaction in early-stage design evaluations was strengthened by its strong alignment with experimental data. The reduced model maintains adequate accuracy, as evidenced by the small (1.2% or less) difference in stress and deflection values between the experimental and model results.
- These findings demonstrate that while the current wing design may satisfy DUL standards, it might be advantageous to add more material in high-stress areas to increase safety margins.
- This research serves as a foundation for design optimization, which could result in weight reductions while maintaining the PC9 wing's structural integrity.
- In conclusion, this FE analysis shows how useful virtual simulations are for aircraft design, providing important information about load-bearing capability and potential structural improvement areas. The verified model contributes to the overall safety and effectiveness of aeronautical structures by offering a dependable tool for iterative design improvement.

7. Composites

Composite materials are increasingly important in aerospace structures due to the growing need for fuel efficiency and performance in an era where oil prices fluctuate globally. Composites, composed of two or more materials that synergize to enhance structural properties, offer a high strength-to-weight ratio that makes them an ideal substitute for traditional aluminium alloys. While aluminium remains a cost-effective choice, composites provide significant long-term benefits. Their reduced weight directly lowers fuel consumption and improves overall operating efficiency, crucial for aircraft manufacturers aiming to minimize costs. The Boeing 707 in the 1950s marked one of the earliest uses of composites, and this material innovation has evolved substantially over time. Today, Boeing's 787 Dreamliner and the Airbus A350 showcase extensive use of composites, with nearly half of their structures composed of composite materials, resulting in lighter, more efficient, and durable aircraft (Spartan, 2020).

The intrinsic properties of composites contribute significantly to their aerospace applications. Their high strength-to-weight ratio helps reduce aircraft weight while ensuring structural integrity, which is crucial for modern aircraft. This weight reduction not only improves fuel efficiency but also extends the range and increases payload capacity. Composites are also highly resistant to corrosion and fatigue compared to metals, reducing maintenance needs and enhancing longevity. Additionally, composites allow for greater design flexibility, enabling engineers to mold complex aerodynamic profiles that maximize fuel efficiency and load-bearing configurations. Their thermal stability further supports performance under diverse temperature conditions, making them suitable for high-performance applications (Miracle, 2015).

7.1 Advancements in modern technologies

State-of-the-art manufacturing techniques have enhanced the use of composites in aerospace. Automated Fiber Placement (AFP) and Automated Tape Laying (ATL) are two cutting-edge methods that allow precise layering of composite plies, ensuring consistency and reducing production time. These techniques also enable tailored fiber orientations, optimizing the structure's strength in critical areas while maintaining lightness. AFP and ATL are widely adopted for producing large parts like wings and fuselage sections, where both weight and strength are paramount. This technological advancement allows aircraft to be constructed with a focus on performance and efficiency (Soutis, 2005).

Advancements in resin and fiber technologies have also significantly improved the capabilities of composite materials. High-performance fibres, such as IM7, T800, and the IM7/977-3 used in PC-9 trainer aircraft wings, deliver exceptional strength and stiffness, meeting the demanding requirements of aerospace applications. Thermoset and thermoplastic matrices offer enhanced toughness, impact resistance, and thermal stability, allowing composites to be used in more critical and structurally demanding applications. Hybrid composites, which combine fibres like carbon and glass, provide additional strength and cost savings. This type of composite is especially useful for components exposed to high impact or vibration, such as fuselage skins and landing gear (Vinson & Sierakowski, 2002).

In recent years, Structural Health Monitoring (SHM) has gained importance, embedding sensors in composite materials to monitor structural integrity in real-time. This approach enables predictive maintenance, detecting micro-cracks or other damage early in the component's life cycle. Integrating SHM in composite aircraft structures represents a shift toward more sophisticated, responsive maintenance practices, enhancing safety and reducing downtime (Giurgiutiu, 2008). Additionally, additive manufacturing techniques have allowed for 3D printing of fiber-reinforced composites, enabling the rapid production of complex parts with customized fiber orientations. This innovation holds transformative potential for aerospace, as it combines lightweight design with enhanced flexibility, paving the way for even more efficient manufacturing processes (Goh et al., 2020).

The environmental impact of composite materials has also led to research into recyclable options. Traditional thermoset composites are challenging to recycle; however, thermoplastic-based composites are gaining attention for their ability to be melted and reformed. This property aligns with the aerospace industry's goals for sustainability, as these recyclable composites can minimize waste. The Boeing 787 and Airbus A350 exemplify the benefits of such advancements, with their composite structures contributing to reduced maintenance needs and considerable fuel savings over their operational lives (Freitag et al., 2018).

7.2 Selected material

We select IM7/977-3 composite as our material for redesigning the wing. IM7/977-3 is a carbon-fiber-reinforced polymer composite widely recognized for its effectiveness in aerospace applications, particularly in critical components like aircraft wings. This composite combines IM7 carbon fibers, which are known for their high tensile strength and stiffness, with a 977-3 epoxy matrix, resulting in excellent durability, impact resistance, and thermal stability. Together, these properties make it particularly suited for demanding applications in aerospace structures (Miracle, 2015). For the PC-9 wing, IM7/977-3 offers numerous advantages over

traditional materials such as aluminum alloys by contributing to substantial weight reduction and enhancing the aircraft's overall performance.

One primary advantage of IM7/977-3 for the PC-9 wing lies in its strength-to-weight ratio. IM7 fibres are designed to achieve high tensile strength, enabling the construction of a lightweight yet robust wing structure capable of withstanding significant aerodynamic loads. This weight reduction is essential in aircraft design as it improves fuel efficiency and increases payload capacity without sacrificing structural integrity (Soutis, 2005). The epoxy matrix, 977-3, also provides excellent environmental resistance, allowing the composite to perform reliably under conditions like high altitude and temperature variation. For trainer aircraft like the PC-9, which benefit from reduced weight and enhanced maneuverability, IM7/977-3 is a highly effective material choice.

By contrast, aluminium 2124 T851 is a high-strength aerospace-grade alloy traditionally valued for its toughness and machinability. However, it lacks the weight-saving benefits offered by IM7/977-3. The higher density of aluminium means more material is needed to achieve equivalent strength, resulting in a heavier structure (Bakis et al., 2002). Although aluminium 2124 T851 provides reliable structural support, its inability to match the lightweight nature of composites makes it less suitable for applications where mass reduction is essential.

Another notable benefit of IM7/977-3 is its enhanced fatigue and corrosion resistance compared to metals. Carbon-fiber composites inherently resist fatigue and corrosion better than aluminum, thus reducing maintenance needs and prolonging the life span of the structure. In contrast, aluminum is susceptible to corrosion in the presence of moisture and oxygen, necessitating protective coatings and routine maintenance, which add to operational costs. This superior fatigue resistance is essential for trainer aircraft like the PC-9, which undergo frequent loading cycles, making durability and reduced maintenance essential for operational efficiency (Vinson & Sierakowski, 2002).

Thermal stability is another critical advantage of IM7/977-3. This composite maintains stable mechanical properties over a wide range of temperatures, which is crucial in high-performance aircraft that experience rapid altitude changes and fluctuating temperatures. Aluminum 2124 T851, while stable at moderate temperatures, can undergo strength degradation at higher temperatures, which may impact performance in extreme conditions (Giurgiutiu, 2008). The enhanced thermal resilience of IM7/977-3 ensures its reliability in temperature-varying environments, which is advantageous for aerospace components exposed to these conditions.

Manufacturing flexibility is another strong point of IM7/977-3, as it can be molded into complex shapes and tailored for specific load requirements using advanced fabrication methods, such as Automated Fiber Placement (AFP) and Automated Tape Laying (ATL). These techniques allow engineers to tailor the fiber orientations to meet specific structural needs, optimizing both aerodynamic and load-bearing characteristics (Freitag et al., 2018). Aluminum 2124 T851, although machinable, does not offer the same level of customization, limiting its suitability for highly optimized structural applications in aircraft design.

In summary, IM7/977-3 proves to be an ideal material for the PC-9 wing due to its lightweight, high strength-to-weight ratio, superior fatigue and corrosion resistance, thermal stability, and design flexibility. In contrast, aluminium 2124 T851, while robust, is heavier and less adaptable, requiring more material and maintenance to achieve similar performance. These advantages of IM7/977-3 translate to enhanced fuel efficiency, lower operating costs, and

reduced maintenance, aligning with the demands of modern aerospace applications focused on maximizing performance and durability.

E1 (Mpa)	164000
E2 (Mpa)	9860
E3 (Mpa)	9860
V12, V13	0.33
V23	0.34
G12, G13 (Mpa)	4950
G23 (Mpa)	2944
Thickness (mm)	0.13
Density (kg/m ³)	1615

Table 15 : IM7/977-3 material properties

7.3 Design Skin

The skin of the wing the most important aspect of the plane as it provides the lift, housing for avionic and flight characteristic of the entire plane. When designing the composite wing, we will refer to the maximum stress of the original aluminum skin in three directions to design the number of layers and orientation of the composite material of the wing while ensuring that the stress value of the wing remains unchanged. To design composites, we will follow the following design strategy:

1. Calculation of force intensity in each orientation:

We calculate force intensity in longitudinal (x) transverse (y) and shear direction. We calculate load intensity by multiplying the DUL stresses from finite element analysis.

Equation 44

$$\sigma_x = \frac{Px}{A} = \frac{Px}{wx * t_{skin}}$$

Equation 45

$$Nx = \sigma_x * t_{skin} \left(\frac{N}{mm} \right)$$

Equation 46

$$Ny = \sigma_y * t_{skin} \left(\frac{N}{mm} \right)$$

Equation 47

$$Nxy = \tau * t_{skin} \left(\frac{N}{mm} \right)$$

In designing composite skins for aerospace applications, orientations of 0, 90, and 45 degrees are commonly selected to optimize the skin's structural performance under multidirectional loads. The 0-degree fibres align with the primary load direction, providing high tensile and compressive strength, while the 90-degree fibres add stability perpendicular to this load, enhancing rigidity and resistance to deformation. The 45-degree fibres contribute to shear strength and impact resistance, helping the structure to better absorb torsional loads and reduce stress concentrations. By combining these orientations, the composite skin achieves a balanced, isotropic-like behaviour, which ensures durability, load distribution, and overall structural integrity under complex loading conditions.

The inclusion of 45-degree fibres in composite skin designs is essential for resisting shear stresses that occur due to aerodynamic and structural loads. When an aircraft is in flight, its skin is subject to a range of forces, including aerodynamic pressure, bending, and torsion. The aerodynamic forces acting on the wings and fuselage generate torsional loads that create shear stress across the skin. This shear stress is particularly significant in areas where the skin experiences twisting or bending forces, such as around the wing roots or fuselage joints. By aligning fibres at 45 degrees, the composite skin can effectively counteract these shear forces, improving torsional stiffness and preventing material deformation or failure. This orientation ensures that the skin can distribute and withstand the multi-directional stresses it faces in flight, enhancing overall structural resilience and performance.

A combination of drag and lift generation creates complex flow patterns over the wing, resulting in shear stresses that the wing skin must withstand. These stresses are particularly critical in composite materials, where fibre orientation and material properties must be designed to effectively counteract the induced shear forces while maintaining structural integrity and performance.

Calculation of Ultimate strengths in each direction:

Following the same principle as above we calculate strengths in each orientation by using hook's law and multiplying young modulus with respective strains for tension or compression

Equation 48

$$Xt^0 = E1 * \epsilon_{ult} \text{ mPa}$$

Equation 49

$$Yt^{90} = E2 * \epsilon_{ult} \text{ mPa}$$

2. Design rules for aircraft structures:

- Minimum of 10% plies in each direction to provide minimum acceptable strength in each direction
- No more than 4 unidirectional plies in each orientation to prevent splitting from cure temperature.
- Balanced and symmetric to avoid warping
- +/- 45-degree plies on outside surface of shear panels to increase resistance against buckling (rib design).

Based on the FEA of wing under design ultimate load cases, we find the stresses in each direction to be positive hence we will use tensile strain to calculate the strengths in each direction. 16

$Sx \text{ (Mpa)}$	125.18
$Sy \text{ (Mpa)}$	100
$Sxy \text{ (Mpa)}$	39

Table 17 stress values in each direction (DUL)

We apply a safety factor of 1.5 to the above stresses:

$S_x (Mpa)$	187.77
$S_y (Mpa)$	150
$S_{xy} (Mpa)$	58.5

Skin thickness is provided to be around 0.8mm and hence we will find load intensities in each direction, based on table 12 and equation 33, 34 and 35,

$$\sigma_x = \frac{Px}{A} = \frac{Px}{wx * t_{skin}}$$

$$Nx = \sigma_x * t_{skin} = 187.7 * 0.8 = 150.216 \frac{N}{mm}$$

$$Ny = \sigma_y * t_{skin} = 150 * 0.8 = 120 \frac{N}{mm}$$

$$N_{xy} = \sigma_{xy} * t_{skin} = 58.5 * 0.8 = 46.8 \frac{N}{mm}$$

Table 18 load intensity in each direction

Nx	150.216	N/mm
Ny	120	N/mm
Nxy	46.8	N/mm

Tensile strains are assumed to be between 4000-5000 $\frac{\mu mm}{mm}$ maximum hence we will determine strengths in each orientation

Ultimate strengths in each orientation can be simplified using following assumptions

$$Xt^0 = E1 * \epsilon_{ult} \text{ mPa}$$

$$Xt^{90} = Xt^{45} = 0.1 * E1 * \epsilon_{ult} \text{ mPa}$$

$$Yt^{90} = Xt^0$$

$$Yt^{45} = Yt^0 = 0.1 * Xt^0$$

$$S^{90} = S^0 = 0$$

Hence the strengths are:

orientation	Strength(mPa)
Xt^0	820
Xt^{90}	82
Xt^{45}	82
Yt^{90}	820
Yt^0	82
Yt^{45}	82
S^0	0
S^{90}	0
S^{45}	250

Table 19 Strength in each orientation

Only assumption we make here is 4 plies in 45 degree orientation as discussed above and then iterate the following.

Based on the above we can apply the Classical Laminate Theory:

Equation 50

$$N_x = N_0 * t_{ply} * X t^0 + N_{90} * t_{ply} * X t^{90} + N_{45} * t_{ply} * X t^{45}$$

Equation 51

$$N_y = N_0 * t_{ply} * Y t^0 + N_{90} * t_{ply} * Y t^{90} + N_{45} * t_{ply} * Y t^{45}$$

Equation 52

$$N_{xy} = N_{45} * t_{ply} * S^{45}$$

Based on these equations:

Table 20 Skin design

orientation	# plies	%plies in each orientation
0	2	25
90	2	25
45	4	50

Our ply configuration would be: $[-45/0/90/45]_s$

Hence total number of plies is 8 hence total thickness is 1.04 mm but the density of composite being very low it will reduce the weight of the wing significantly. Hence to withstand the same number of forces as aluminum (original) wing, the composite should have a total thickness of 1.04 mm.

7.4 Main Spar design

The redesign of the main spar using composite materials aims to achieve significant mass reduction while maintaining equivalent mechanical properties to the original aluminum spar. Given the critical role of the main spar in carrying bending stress in the wing, the transition to composite materials requires careful consideration. To fully assess this, two cases are explored in this study.

In **Case 1**, the composite spar is designed with the same dimensions as the original aluminum spar. This case focuses on calculating the required number of composite layers and their orientation to match the aluminum spar's stiffness and strength while achieving a reduction in mass. The primary objective here is to validate the feasibility of using composite materials to reduce weight, and to provide a foundational rationale for why composites are beneficial in spar and overall wing design.

Case 2 shifts focus toward a more application-based engineering approach by designing the composite spar to meet the bending and axial stiffness of the aluminum spar, rather than retaining the same dimensions. Here, the spar dimensions will be adjusted to best leverage composite properties for practical use, aiming to maintain the same mechanical performance as the aluminum spar. This case emphasizes exploring both the mass savings and any dimensional differences required to match stiffness, which could inform real-world design adjustments.

Design:

In case 1 we will determine the layer of composite by dividing total thickness of spar for each flange and web by ply thickness. We calculate for three points, wing-fuselage junction as this is where a lot of forces and moments concentrate as it can be seen from static analysis as a reference to give us an idea. Next, we would consider wing tip as it faces a lot of deflection. Hence based on those,

Table 21 number of plies at rib 5

dimensions in mm		
Rib 5		
	web	flange
thickness	2.3	17
ply thickness	0.13	0.13
#plies	17.6923077	130.769231

At rib 5 we will need around 18 plies in web and 130 in flange due to the higher thickness requirements and the weight for these will also significantly go down due to the lower aluminum density.

For the tip both would be around 18 plies due to the same thickness overall (2.3mm).

Our final orientation is:

Web: [$\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$]

Flange: [$\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 90° , 0° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 90° , ..., $\pm 45^\circ$]

This orientation setup should provide robust shear resistance, axial stiffness, and secondary load protection while minimizing the risk of delamination or impact damage.

The following equations are used to derive the axial and bending stiffness:

$$EA_{eq} = \sum (EA)_i$$

$$EI_{eq} = \sum (EI)_i$$

hence for Aluminium

$$EA_{eq} = 202986000 \text{ N} \quad EI_{eq} = 2.46342\text{E}+12 \text{ N/mm}^2$$

web		Flanges	
Em(mPa)	73324	Em(mPa)	40889
Eb(mPa)	42444	Eb(mPa)	33735

Table 22 Em Eb values for composite (IM77/973)

	Aluminium	IM77/973	% difference
(EA) _{eq} kN	202986000	102351691.2	49.57697023
(EI) _{eq} Nmm ²	2.46342E+12	1.18719E+12	51.80706185

Table 23 Axial and bending stiffnesses for both Aluminium and Composites

Hence it can be seen that the axial and bending stiffnesses for composites are way less than aluminum when the cross-sectional area is kept the same. These show case that the area of the spar should be significantly increased to fit larger number of composites. This is needed so that the axial stiffness is same for composite and aluminum as compared to the 49% difference. These will be addressed in case 2.

In case 2 we will iterate with the spar dimensions so that the axial stiffness is at par with the aluminum. According to the difference of EA values from table 18, we need to increase the area, this can be done through changing thickness as the width is constrained. As modification of width and height of spar would require redesigning the entire wing which is out of the scope of our project.

Table 24 New dimensions based on Axial stiffness

Location	Width (mm)	Old thickness (mm)	New thickness (mm)	Em(mPa)	Eb(mPa) old	Eb(mPa) new
top	70	17	36	40889	33735	71438.8235
bottom	70	17	36	40889	33735	71438.8235
web	226	2.3	4.6	73324	42444	84888

Initially as calculated based on the original thickness we had a significant difference in the axial stiffness matrix as seen above hence we iterated to around twice the original length to give us the similar values as axial stiffness as aluminium and a greater bending stiffness. The results are further depicted in table 20.

Table 25 Stiffness comparison after iteration.

	Aluminium	IM77/973	% difference
(EA)eq kN	202986000	202976760.5	0
(EI)eq Nmm ²	2.46342E+12	2.52838E+12	-2.63714629

Hence, we can derive the number of plies to 278 for top and bottom flanges and 36 for web.

Final layer-up:

Flange Layup (278 Plies)

For the flange, we will:

Use 0° plies along the beam axis to maximize bending stiffness, positioning them farther from the neutral axis.

Include ±45° plies for shear resistance and impact durability, especially on the outer layers.

Add 90° plies to protect against secondary load cases and enhance stability.

Ply Orientation Breakdown for Flange:

0° plies: 30% (83 plies)

±45° plies: 40% (111 plies)

90° plies: 20% (56 plies)

Additional ±45° plies for impact resistance: 10% (28 plies)

This distribution ensures a balanced and practical layup to handle axial and shear loads effectively.

Suggested Flange Layup Sequence:

[$\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 90° , ...]

This sequence can be repeated while ensuring symmetry, with $\pm 45^\circ$ plies on the outermost layers to enhance impact resistance. For the total of 278 plies, continue the above pattern, ensuring an equal distribution of 0° , 90° , and $\pm 45^\circ$ plies, as outlined.

Web Layup (36 Plies)

For the web, the primary focus will be on shear resistance, with additional plies at 0° and 90° for structural stability and load distribution.

Ply Orientation Breakdown for Web:

$\pm 45^\circ$ plies: 50% (18 plies) for shear resistance.

0° plies: 25% (9 plies) along the beam axis.

90° plies: 25% (9 plies) for stability and secondary load cases.

Suggested Web Layup Sequence:

[$\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 90° , $\pm 45^\circ$, 0° , $\pm 45^\circ$, 90° , $\pm 45^\circ$, 0° , ...]

This sequence maintains a balance of shear strength, axial stiffness, and stability for a total of 36 plies.

Weigh saving:

Hence, we will now proceed to validate our model in FEM.

FEM RESULTS AND VALIDATION

In the FE analysis the same wing structure was made to undergo the static structural analysis, but the newly selected composite i.e., IM7/977-3 was used as the material for the whole model. The goal of this analyses was to generate the parameters like deflection, equivalent stress, principal stress, Principal stress, Principal strain, etc. More pictures of analysis can be found on [appendix](#)

The findings for the three cases performed are in the table below

PARAMETER	Case1	Case2	Case3
Load (N/mm)	104.35	132	187
Maximum deflection(m)	0.19525	0.25635	0.34055
Minimum deflection(m)	-0.0015342	-0.002014	-0.002676
Average deflection(m)	0.07187	0.094359	0.12535
Equivalent (Vonmises) stress (Pa)	197.85	259.76	345.08
Equivalent elastic strain(mm/mm)	11.4155e03	1.8583e-03	2.4687e-03
Principal stress(Pa)	109.75	144.09	191.41

Principal Strain(mm/mm)	8.9255e-04	1.1718e-03	1.5567e-03
Shear stress(Pa)	1.8106e+05	2.3772e+05	3.158e+05
Shear Strain	2.9589e-06	3.8847e-06	5.1606e-06

Comparison of Results

The main objective of this project is to find how the material and design change influences the object response to loads. Therefore, the results from the aluminium cases are compared to that in composites.

PARAMETER	CASE1(104.35 N/mm)		CASE2 (137 N/mm)		CASE 3 (182 N/mm)	
Material	Al-2124-T851	IM7/977-3	Al-2124-T851	IM7/977-3	Al-2124-T851	IM7/977-3
Deflection(m)	0.49474	0.07187	0.64954	0.094359	0.86289	0.12535
Equivalent (Vonmises) stress (MPa)	579.5	197.85	760.81	259.76	1010.7	345.08
Principal stress(MPa)	317.56	109.75	416.93	144.09	553.87	191.41
Principal Strain(mm/mm)	5.7851e-03	8.9255e-04	7.5952e-03	1.1718e-03	1.009e-02	1.5567e-03
Shear stress(Pa)	1.2006e+06	1.8106e+05	1.5765e+06	2.3772e+05	2.0943e+06	3.158e+05
Shear Strain	4.3694e-05	2.9589e-06	5.7366e-05	3.8847e-06	7.6209e-05	5.1606e-06

Discussion

Stress and strain comparison

When comparing the results, we can see that the aluminium alloy is experiencing higher maximum stresses and strains in each cases. Which portrays that aluminium can withstand extreme loads, since it has higher deformation under those loads

But the composite material shows much more lower values of stresses and strains, which portrays that the composite material distributes the load more effectively and lower strain levels are maintained.

Deflection

The deflection in aluminium is so high across all the cases, the discussion regarding this was made before in FE analysis of Aluminium. Here since the skin thickness is greater and the spar has been modified, the deflections are way lower than aluminium. The deflection for the aluminium in case 3 was 0.86829m whereas the deflection for the same load in composite is only 0.12535m. This explains that the composites possess a higher stiffness-to-weight ratio, which is depicted in reduction of deflection.

After the comparison of result, it is evident that IM7/977-3 performs better than Al2124-T851. The composite works better in the terms of reducing deflection and maintaining required strength under comparable loads. Which is a major criterion to prove that composites are more favourable to be used for aerospace applications. Aluminium can withstand higher stresses, which can lead to structural challenges in elevated loads. Which again makes us to go towards composite materials because of its lighter weight and resilient structure.

Sustainability and Integration

Sustainability is a critically important goal for human activity and development. Sustainable development is increasingly becoming a goal to which numerous countries throughout the world aspire and it has three distinct components: environmental sustainability, economic sustainability and social sustainability and achieving the balance between these three factors is a very difficult task. Engineering sustainability is viewed as a broad concept that encompasses the responsible use of engineering within various systems. These systems involve processes and technologies for resource extraction, transformation into usable forms, transportation and storage, as well as the application of engineering products and processes to deliver valuable services (Rosen, M.A., 2012)

Life Cycle Assessment

The objective of this life cycle assessment is to evaluate the environmental impact of a composite wing compared to an aluminum wing. In this assessment, there are four key aspects that will be assessed:

- Energy consumption during manufacturing
- Raw material sourcing
- Emissions during production and operation
- End-of-life disposal or recycling options

Energy consumption during manufacturing

- Composites: The energy consumption can vary depending on the condition but in this case for IM7/977-3, the cure temperature is 177 degrees Celsius (Miller, S.G., Sutter, J.K., Scheiman, D.A., Maryanski, M. and Schlea, M., 2010) which requires a substantial amount of energy input during the manufacturing process.

Raw material sourcing

Carbon Fiber (IM7)

- IM7 fibre is manufactured through a multi-step process that involves precursor fibre production

Epoxy Resin (977-3)

- This involves chemical reactions and majority of the production of resins is based on the reactions between BPA and epichlorohydrin in the presence of a basic catalyst and are highly reactive (Saba, N., Jawaidd, M., Alothman, O.Y., Paridah, M.T. and Hassan, A., 2016)

Emissions during production

- IM7 Carbon Fiber produces 54 kg of CO₂
- Epoxy Resin (977-3) produces 2.25 kg of CO₂

End-of-life disposal or recycling options

- Mechanical Recycling
- Chemical Recycling
- Landfill

Trade-off Analysis

Category	IM7 Carbon Fiber	Bio-composite
Key Properties and Data		
Tensile Strength	5670 MPa	23 MPa
Density	1780 kg/m ³	928 kg/m ³
Production Energy	1100 MJ	203 MJ
Emissions	54 kg of CO ₂	12 kg of CO ₂
Environmental Benefit Analysis		
Volume of the Wing	26.17 m ³	26.17 m ³
Wing weight	46582.6 kg	24285.76 kg

Alternative Manufacturing Methods		
Manufacturing Options	Traditional Layup	Automated Fiber Placement (AFP)
Key Properties and Data		
Energy Consumption	500 MJ per wing	300 MJ per wing
Waste Generation	20%	5%
Emissions	10 kg CO ₂ e per wing	5 kg CO ₂ e per wing
Environmental Benefits Analysis		
Energy Consumption	500 MJ	300 MJ
Waste Generation	9316.52 kg	1214.288 kg
Emissions	10 kg of CO ₂	5 kg of CO ₂

Material and Manufacturing Method Selection

The selected material for the wing is bio composite as it reduces the weight of the wing by 22,296.84 kg improving fuel efficiency. On top of this, bio composite also has a much lower energy production and has much lower emissions compared to IM7 carbon fibre. For manufacturing, automated fibre placement is more beneficial compared to the traditional layup as it has reduced energy consumption, lower waste generation and reduced emissions.

Human Factors and Usability Assessment

Maintenance:

Inspect ability

- **Current State:** Technicians currently rely on visual inspections that require scaffolding or ladders for full access.
- **Proposed Design:** Integrate inspection ports made from transparent composite materials at key structural points. This allows technicians to conduct visual checks without additional equipment making it more efficient

Repairability

- **Current State:** Repairs often require specialized skills and materials, making it lengthy.

- **Proposed Design:** Create standardized repair kits that includes pre-measured patches and resin for quick application. Implement training sessions to familiarize technicians with these kits.

Replaceability

- **Current State:** Full wing replacements require extensive disassembly of the aircraft.
- **Proposed Design:** Develop a modular wing system where sections can be detached and replaced independently. Use an interlocking mechanism to simplify reattachment.

Pilot Interaction:

Handling and Performance

- **Current State:** Pilots report varying handling characteristics during different flight manoeuvres, making it difficult to predict aircraft behaviour.
- **Proposed Design:** Use flight simulators to refine wing design based on pilot feedback. Adjust wing profiles to ensure more predictable handling, especially in critical phases of flight.

Feedback Mechanisms

- **Current State:** Limited real-time data is available regarding wing performance during flight.
- **Proposed Design:** Integrate an advanced avionics system that provides real-time data on wing loads and performance metrics, allowing pilots to adjust their flying style accordingly.

Safety:

Usability Risks

- **Current State:** Emergency procedures are often not intuitive, leading to confusion during high-stress situations.
- **Proposed Design:** Create an easy-to-follow, color-coded emergency procedure guide specifically for wing-related issues, enhancing quick decision-making.

Ergonomics

- **Current State:** Control systems are not optimally placed, leading to potential strain during long flights.
- **Proposed Design:** Redesign control placement based on ergonomic studies, ensuring all controls are within easy reach and require minimal effort to operate.

Seamless Integration with Aircraft Systems

Step 1: Develop Strategies to Interface New Wing Designs with Legacy Systems

1. Comprehensive Systems Assessment

Conduct a detailed evaluation of the existing avionics, hydraulic, and control systems in the PC-9 to identify interfaces that will require adaptation.

2. Modular Design Approach

Design the new wing with a modular interface that allows for easy connection to existing systems. This includes plug-and-play connectors for avionics and standardized hydraulic fittings.

3. Standardized Communication Protocols

Implement standardized communication protocols such as ARINC 429 allows data from the wing's sensors to be transmitted to the aircraft's avionics.

4. Simulation and Testing

Use computer-aided design (CAD) tools and flight simulators to test the integration of the new wing with the legacy systems before physical implementation.

Potential Integration Challenges:

1. Challenge: Retrofit Compatibility with Older Systems

Proposed Solution:

Develop retrofitting kits that include adapters and bridging components to connect new sensors and systems with existing wiring harnesses. If the new wing includes advanced sensors that require different electrical connections, the kit could provide standardized adapters that enable them to interface with the legacy avionics.

2. Challenge: Ensuring Reliability Without Disruption

Proposed Solution:

Incorporate redundant systems for critical functions, such as flight control surfaces. Implement dual hydraulic systems where one set operates the new wing controls while a secondary set acts as a backup, ensuring continued operation in the event of a failure.

3. Challenge: Training and Familiarization

Proposed Solution:

Develop training programs and simulations that focus on the integration of the new wing systems, emphasizing how they interact with legacy systems. Create a series of interactive training modules that allow pilots and maintenance personnel to practice using the new systems in simulated environments, ensuring they are prepared for real-world operations.

4. Challenge: Monitoring Performance Post-Integration

Proposed Solution:

Install advanced monitoring systems that continuously assess the performance of the new wing in relation to legacy systems. Use onboard diagnostics that alert pilots to any discrepancies between the expected performance of the new wing and actual readings, allowing for immediate corrective actions.

Conclusion

The comprehensive analysis of the PC-9 wing structure highlighted critical aspects of its performance, focusing on safety, resilience, and efficiency. By examining the lift forces, fracture resistance, fatigue life, dynamic response, and the benefits of composite materials, the study provided a clear understanding of the structural integrity and reliability of the wing. These analyses yielded crucial findings regarding the distribution of stresses, the margin of safety across various structural elements, and the feasibility of implementing advanced composite materials. In this comprehensive study, a sustainable approach to designing the PC-9 aircraft's wing was explored, focusing on lightweight, durable composite materials to replace traditional aluminium. Emphasizing sustainability, the redesign incorporated considerations of environmental, economic, and social aspects of engineering. The project sought to reduce the

aircraft's carbon footprint, enhance efficiency, and improve resilience by utilizing IM7/977-3 composite material. This approach aligns with modern engineering sustainability principles, which advocate for responsible resource use, minimized environmental impact, and efficient manufacturing processes that extend product life cycles. Through Life Cycle Assessment (LCA), the environmental impact of the composite wing was evaluated across four critical dimensions: energy consumption, raw material sourcing, emissions, and end-of-life disposal options. Compared to traditional aluminium, the IM7/977-3 composite required higher energy input during manufacturing due to the high curing temperature. However, significant advantages were identified in emissions and waste management. With carbon fibre emissions at 54 kg of CO₂ per unit and epoxy resin at 2.25 kg, the composite alternative presented lower overall greenhouse gas emissions during production. Additionally, recycling methods, including mechanical and chemical recycling, offer viable options for the composite wing's end-of-life, further supporting its sustainable profile. Trade-off analyses and alternative manufacturing methods also revealed the benefits of using bio-composites and advanced techniques like Automated Fiber Placement (AFP). The selected bio-composite reduced the wing's weight by over 22,000 kg, resulting in substantial fuel efficiency gains. Manufacturing through AFP not only lowered energy consumption but also reduced waste generation from 20% to 5%, reinforcing the environmental benefits of bio-composites over conventional materials. Additionally, AFP's precision layering minimized emissions, yielding a cleaner production process compared to traditional methods. The sustainability-driven redesign included human factors and usability improvements, aimed at enhancing maintenance, repair, and usability. The proposed inspection ports and modular wing design reduced maintenance complexity, while standardized repair kits streamlined the repair process, reducing downtime. In addition, advanced avionics were integrated to monitor real-time wing performance, enabling pilots to make more informed adjustments mid-flight. These ergonomic considerations support operational safety, simplify emergency procedures, and improve pilot and maintenance crew interaction with the aircraft systems. Furthermore, a seamless integration strategy was proposed for interfacing the new wing with the PC-9's legacy systems. By implementing a modular design, standardized communication protocols, and retrofitting kits, the project addresses potential challenges in adapting new systems to older avionics and hydraulics. Redundant systems and advanced monitoring mechanisms were also suggested to enhance the wing's reliability, ensuring the aircraft's operational continuity without disruption. In conclusion, this report presents a holistic and sustainable redesign for the PC-9 wing, prioritizing the environmental, economic, and operational facets of sustainability. The transition from aluminium to IM7/977-3 composites align with long-term goals for reducing emissions, enhancing fuel efficiency, and minimizing resource use. By integrating advanced manufacturing methods, focusing on usability, and ensuring compatibility with existing aircraft systems, the design offers a blueprint for modern, sustainable aviation engineering. The lift and stress analysis confirmed that the wing sustains aerodynamic loads of approximately 17,329 N, with safety standards met through a 1.5 safety factor in compliance with FAR 23 and ASTM guidelines. Using the Kutta-Joukowski theorem, we were able to relate lift generation to circulation and consider the impact of vortices, while the elliptical lift distribution closely matched predictions from Prandtl's lifting line theory and software simulations. Resulting direct stresses were measured as 95.89 MPa in compression on the top flange and 83.4 MPa in tension on the bottom flange, which aligns with real-world observations of wing behaviour in-flight. This led to safety margins of 3.8, 3.5, and 10 across different points, showcasing the structural robustness of the wing under operational loads. The fracture analysis delved into three primary crack scenarios—cracks at flange holes, centre cracks on the web, and edge cracks on the flanges—demonstrating that the spar maintains adequate safety margins against brittle fracture and plastic collapse. Cracks emanating from holes (Case 1) showed a lower

safety margin of 2.24, underscoring brittle fracture as the primary concern in such areas. In contrast, the centre cracks on the web (Case 2) had a safety margin of 4.56, indicating a stronger resistance to fracture. Edge cracks on the flange (Case 3) displayed a safety margin of 3.53, which is sufficient but emphasizes the criticality of designing against fracture initiation around high-stress points like holes. These findings support the recommendation to reinforce specific areas to mitigate stress concentrations, thereby improving fracture resistance. Dynamic analysis and fatigue assessment incorporated both safe-life and damage tolerance approaches, providing a robust evaluation of the wing's durability. The safe-life method estimated a maximum cycle count of 10,000 before significant cracking could occur under the given loading spectrum. Meanwhile, damage tolerance offered a secondary safeguard by analysing how cracks might evolve between inspections, providing tailored inspection intervals for different areas: 1,060 cycles for the web, 47,900 flights for flange cracks, and 15,500 flights for edge cracks. This combined approach ensures that the wing remains structurally sound and allows for both preventative maintenance and resilience to undetected damage, thus enhancing the PC-9's reliability in service. The experimental analysis presented displacements under incremental loading, with empirical data at 21.62 mm, 40 mm, and 61.06 mm for corresponding loads. These values, when compared to FEA results, validated the computational model and illustrated how different components react to varying loads. The high tensile and shear strains in the bottom spar underscored the importance of designing for both tensile and shear strengths across critical wing components. Stress and strain distributions also varied across the structure, further affirming the need for a comprehensive design approach that incorporates both static and dynamic load considerations. The examination of composite materials as alternatives to aluminium brought valuable insights into improving the PC-9 wing. IM7/977-3, a carbon-fibre-reinforced polymer, was shown to possess a high strength-to-weight ratio, corrosion resistance, and improved fatigue performance. By comparing axial and bending stiffnesses, it was found that increased thickness was needed to match the aluminium spar's stiffness, but the material's low density offered substantial weight savings. In Case 1, an equivalent ply configuration was achieved to match the aluminium spar's performance, while Case 2 adjusted the spar's dimensions for optimal weight and stiffness balance. These adjustments demonstrated that IM7/977-3 composites can reduce the overall wing weight while achieving the necessary structural strength, making it a viable option for modern aerospace applications. Finally, the composite wing skin and spar were designed with ply orientations of 0° , $\pm 45^\circ$, and 90° to withstand axial, bending, and shear loads. In the main spar, a balanced layup of 278 plies in the flanges and 36 in the web ensured robust load-bearing capabilities, impact resistance, and stability under operational stresses. This detailed configuration allowed the wing structure to achieve resilience, improved longevity, and reduced maintenance requirements, aligning well with aerospace engineering goals for modern aircraft design. When the results of composites from FEM was compared to that of aluminium, composite performed great by distributing the load more effectively and it has also influenced in reducing the weight of the wing. In conclusion, the PC-9 wing structural analysis demonstrated that the current design meets safety and operational standards. However, adopting composite materials like IM7/977-3 could enhance the wing's efficiency, reduce weight, and extend its service life. The study's combination of fracture analysis, fatigue assessment, and material redesign offers a comprehensive pathway to optimizing the wing for both performance and safety, ultimately supporting the long-term reliability and efficiency of the PC-9 in service.

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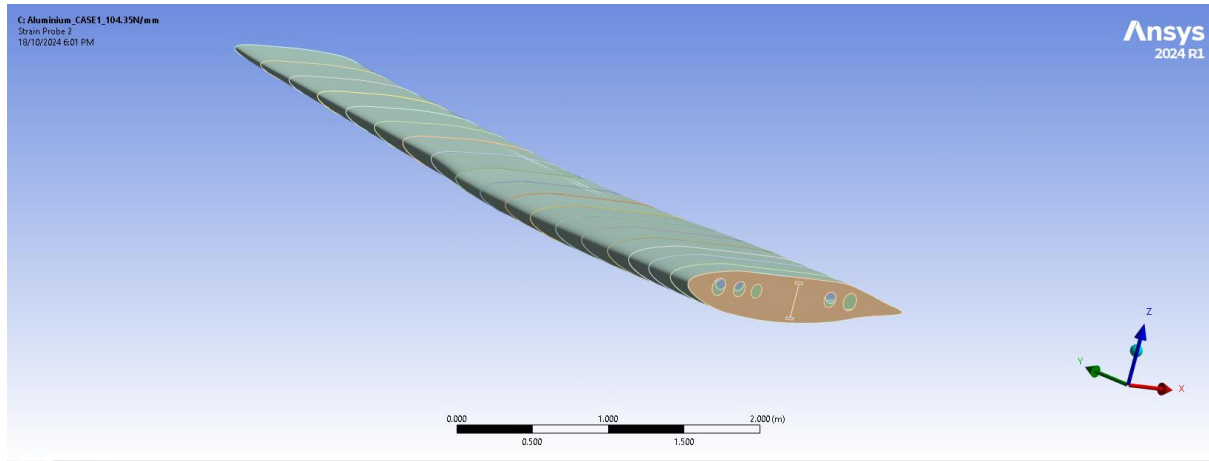
APPENDIX

Margin of safety without the safety factor of 1.5

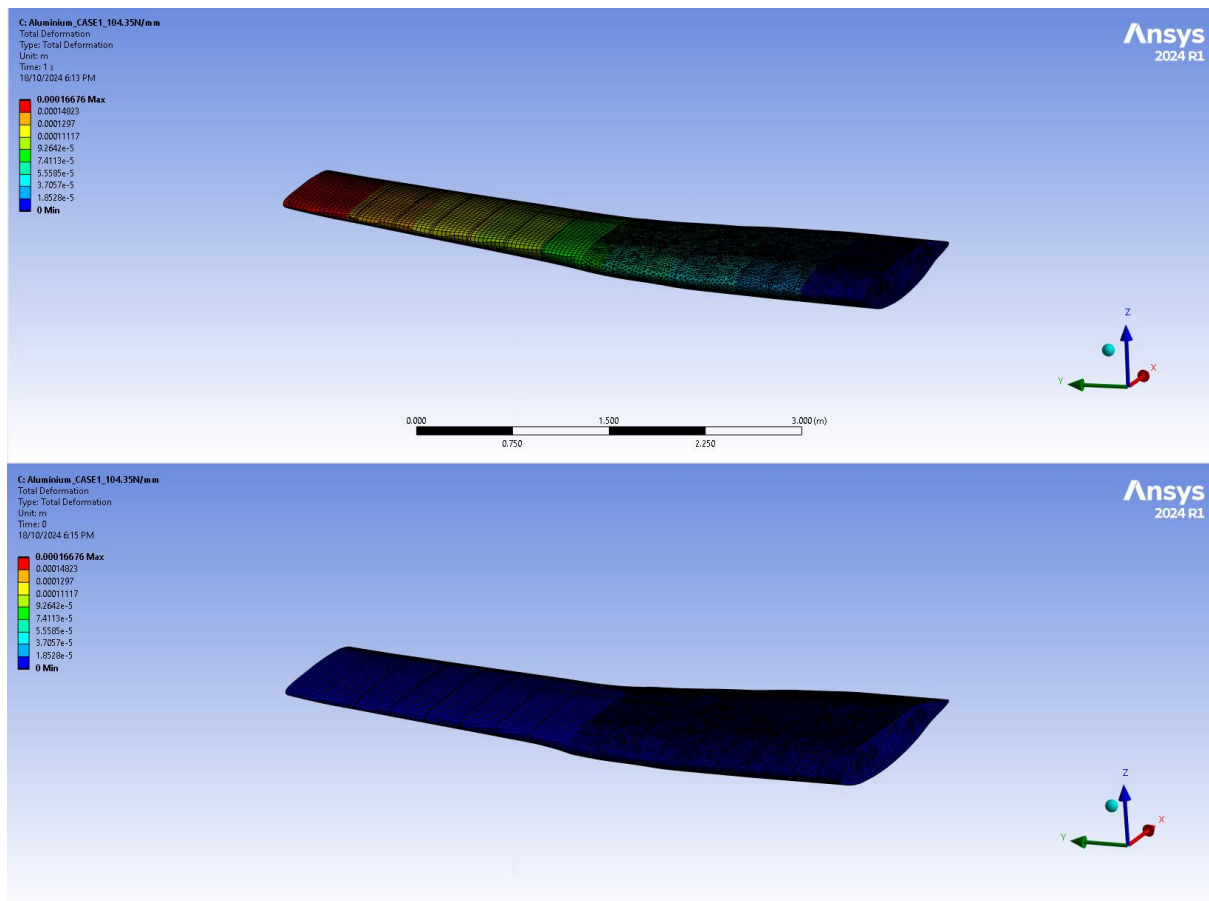
location	point	x(mm)	z(mm)	σ_x	σ_y	q	τ	σ_1	σ_2	Von Mis	MoS	Type
top of flange	A	0	130	-6.4630568	-63.810065	0	2.4053717	-6.3623424	-63.91078	60.9790529	6.23199163	compression
web flange	ju B	-1.15	113	-5.6178878	-55.465672	-49.23790426	-21.407784	2.31386847	-63.397428	64.5854568	5.82816259	compression
centroid	C	0	0	0	0	-49.23790426	-21.407784	21.4077845	-21.407784	37.0793704	10.8934058	compression

Aluminium FE Images

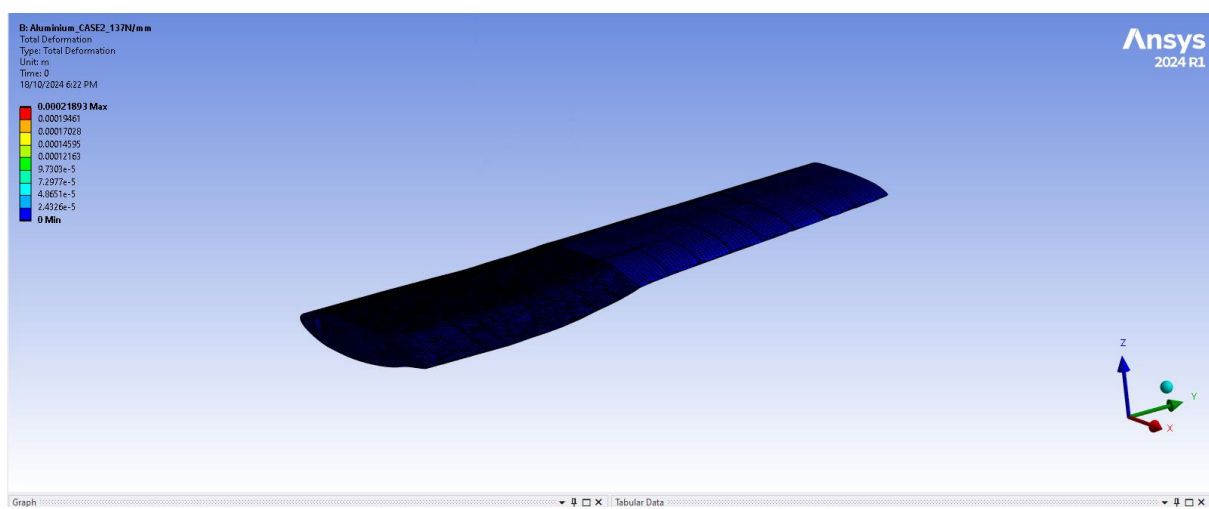
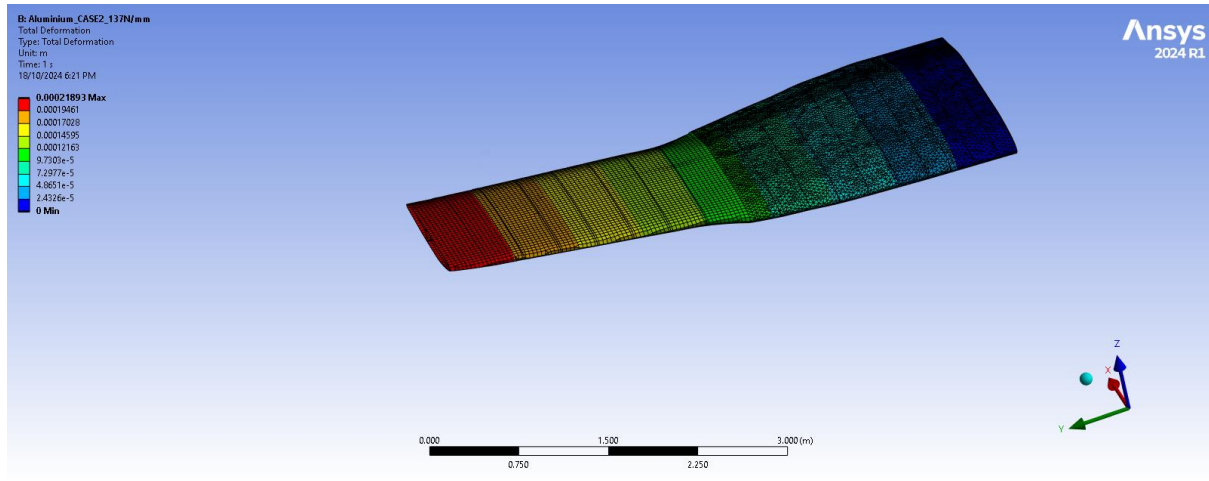
Aluminium model



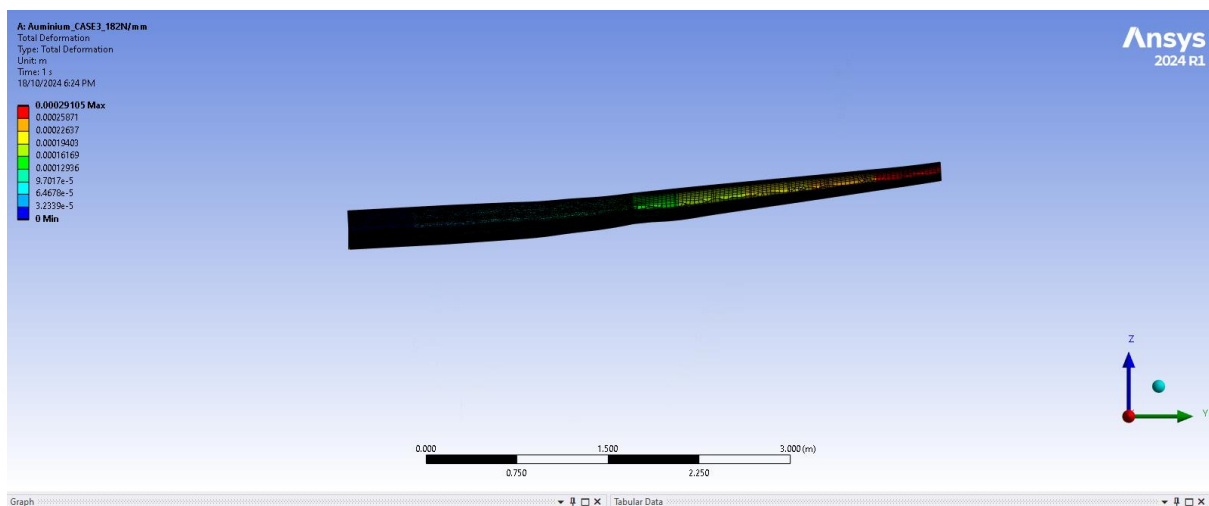
Case-1: Maximum and minimum deflection

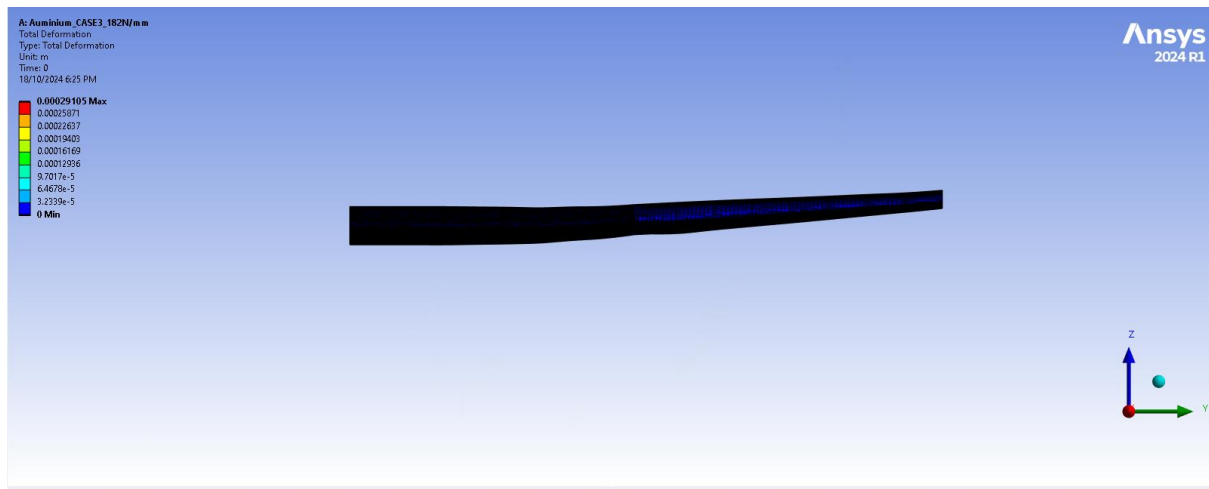


Case-2: Maximum and minimum deflection

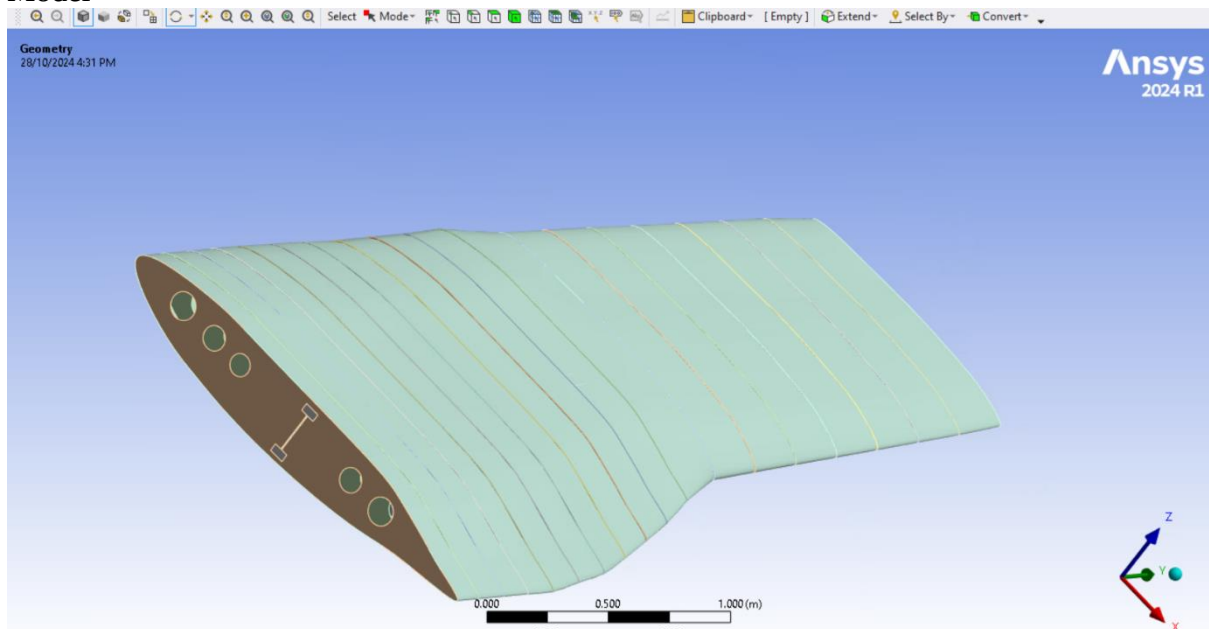


Case-3: Maximum and minimum deflection

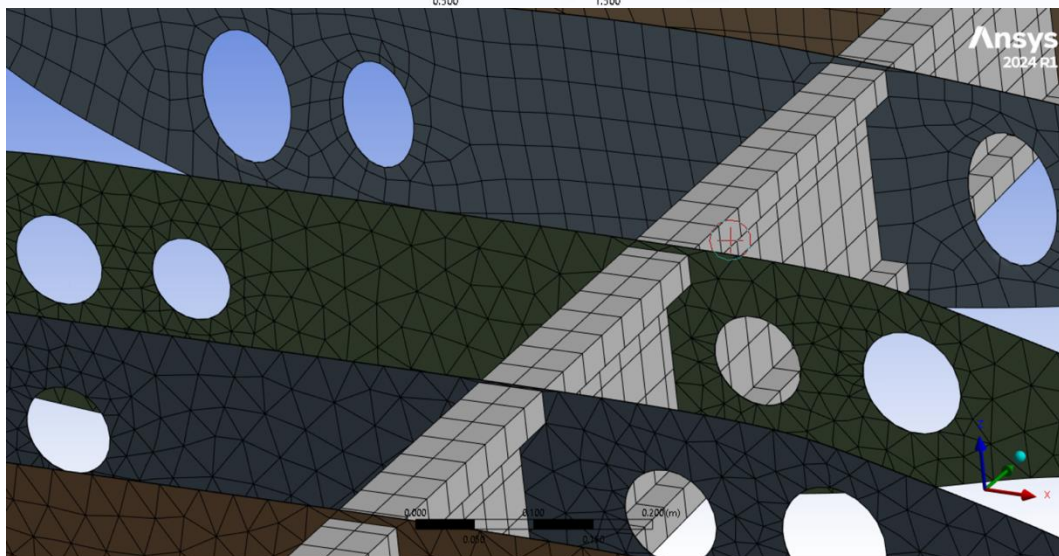
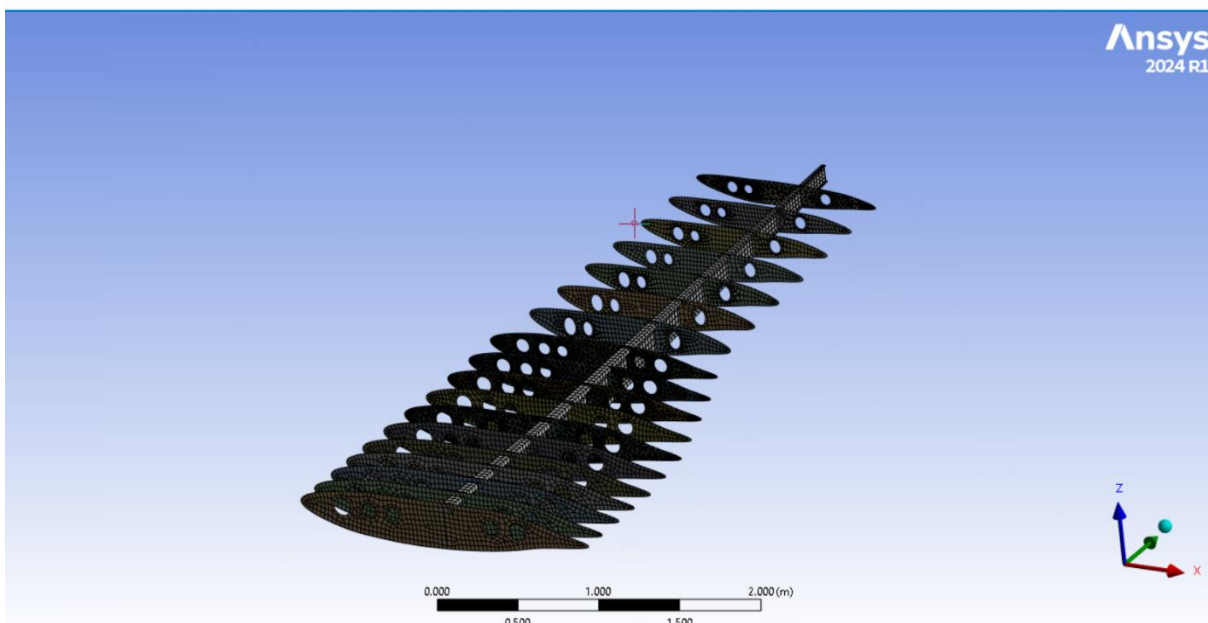
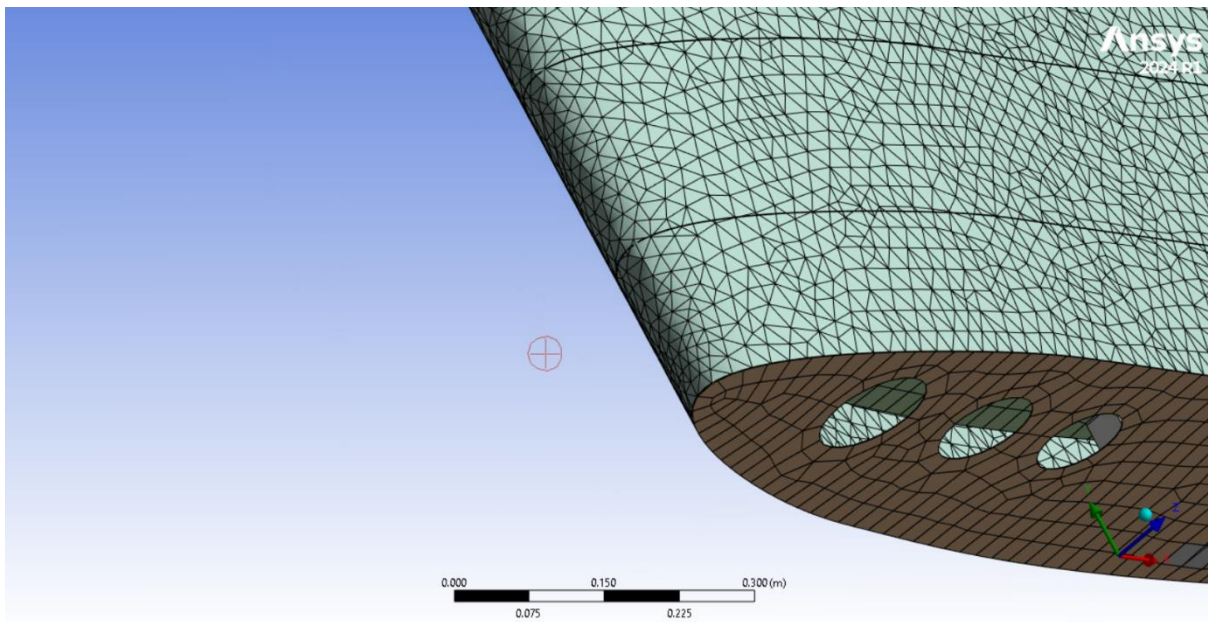




Composite Model



Mesh



Deflection

All cases combined(Minimum and maximum)

